

CRF Complete Edition v17.15 — Part K

ZC83–87: Floor-Lattice Ordering · Complete Sector Synthesis

Observed-Bridge Bracket · Strong Running Bridge · Temporal Continuum Bridge

Four closures: floor lattice ordered, ε_{SC} ontology, Strong running bridge, temporal dt

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If you are just arriving — Part K in plain language

CRF attempts to derive the constants and force structure of physics from a single primitive called Distinction (δ). Earlier Parts built four tiny “instability floors” — thresholds where a force sector tips from one stable state to another — and noticed they sit very close together. **Part K** settles four loose ends about them: it proves *why* the floors fall in the order they do, shows that one observed value is the exact geometric mean of two others, confirms the Strong force runs the same way electromagnetism does, and finally connects CRF’s discrete “ticks” of time to ordinary continuous time.

Minimum vocabulary: δ (*Distinction*) — the one starting idea everything is built from; ε *floors* — four close-but-distinct instability thresholds, one per force sector; *geometric mean* — $\sqrt{ab}$, the kind of average that turns out to relate the floors exactly; *running* — how a coupling changes with scale; *gap* / *GAP*-... — an open question the chain has not yet closed. Full symbol list: Master Notation Key (front matter).

These results are pieces of a map under construction, not a claim of final truth: each carries the conditions under which it would fail (see front matter, “Map, not reality”).

Part LIII

ZC83–84 Cluster: Floor-Lattice Ordering and Complete Sector Synthesis (v17.15)

Part LIII — ZC83–84: The Floor Lattice, Ordered and Synthesised (v17.15)

Part LIII opens the ε -ontology closure arc with two papers: the first orders the complete instability-floor lattice and the second assembles every $\mathbb{Z}_{15}$ sector result into one falsifiable synthesis.

ZC83 (The Floor Lattice Ordering Theorem) proves that the ordering $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$ is forced by $\mathbb{Z}_{15}$ sector-weight arithmetic (THM-ZC83-01), and delivers the central gift FIND-ZC83-01: the bridge floor is the *geometric* mean of the two sector floors, exactly ($\sqrt{\varepsilon_{\text{EM}} \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$), while the arithmetic mean is strictly larger (FIND-ZC83-02). It opens GAP-ZC83-01 (the ontology of the observed bracket $\varepsilon_{\text{SC}} = 0.007605$).

ZC84 (The Complete Sector Synthesis) collects eight simultaneous consequences of δ alone into THM-ZC84-01 — the full four-sector floor lattice, the geometric-mean identity, the mirror symmetry $\Gamma_{\text{EM}} + \Gamma_{\text{strong}} = 2\Gamma_{\text{W}}$, the cosmological lock $w_{\text{DE}} + 3\Delta\Gamma = \Gamma_{\text{CRF}}$, and the uniqueness of $\mathbb{Z}_{15}$ — and presents the four-sector falsification table (COR-ZC84-01) with zero free parameters.

Key results: THM-ZC83-01 (floor lattice ordering, forced); FIND-ZC83-01 (geometric-mean identity, exact); THM-ZC84-01 (eight-fold synthesis); COR-ZC84-01 (four-sector falsification table).

Gap opened: GAP-ZC83-01 (ε_{SC} ontology and closed form) — carried forward to Part LIV.

Phase Misalignments preserved: PM-ZC83-01 (the “is ε_{SC} the arithmetic mean?” probe; pattern: false-pattern test, ruled out exactly); PM-ZC84-SCOPE-01 (synthesis paper — every identity is inherited, none newly derived).

Of all the quantities CRF derives, the four instability floors are the ones that should make a careful reader uneasy. They are numbers — 0.007494, 0.007550, 0.007599, 0.007705 — and they sit so close together that the eye wants to call them the same value measured four times. They are not. Each belongs to a different force sector, each was derived through a different sub-chain, and yet they line up in a strict order with gaps of less than a tenth of a percent between neighbours.

Until this paper, that ordering was an observation, not a result. The floors had been computed one at a time, in separate papers, for separate reasons; nobody had gone back and asked whether their arrangement was forced or merely how things happened to fall out. If it were accidental, the whole picture would be fragile — a different choice upstream could reshuffle them and nothing would forbid it. If it were forced, then the ordering itself carries information about how the sectors relate.

ZC83 settles the question in the second sense, and in doing so turns up something sharper than expected: the middle floor is not merely *between* its neighbours, it is their exact geometric mean. That is the kind of relation that does not happen by accident. The sections below state the ordering as a theorem, exhibit the two-line proof of the mean identity, and record honestly the one quantity this still does not explain — the observed bracket that sits a hair above the rest, which becomes the open question carried into the next cluster.

Abstract

ZC80 collected seven electroweak observables at R_0 ; ZC81 added the Strong sector floor; but the *ordering* of the five ε -floor values had never been proved from first principles. This paper closes that gap.

THM-ZC83-01 (Floor Lattice Ordering, Near-Formal):

$$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}},$$

proved in three steps from the $\mathbb{Z}_{15}$ sector-weight ratio $r = C_W/C_{\text{EM}}$ and the two-layer gap of THM-ZC71-01.

FIND-ZC83-01 (Geometric-Mean Identity, Exact — Gift):

$$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}} \quad \text{exactly.}$$

Two-line algebraic proof: the sector-weight exponents cancel. The bridge floor is the geometric mean of the two sector floors — a consequence of the $\pm 1/4$ exponent structure of LEM-ZC67-01.

FIND-ZC83-02: $\text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) > \varepsilon_{\text{univ}}$ (strict AM–GM; AM exceeds $\varepsilon_{\text{univ}}$ by 0.0096%). **PM-ZC83-01** registers that $\varepsilon_{\text{SC}} = 0.007605$ is neither the AM nor the GM of the defined floor pair, opening **GAP-ZC83-01** (eps_SC ontology).

0.1 Background and Scope

Numerical Summary — ZC83 (T-canonical, R_0)

Quantity	Value	Gap	Source
$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$	0.007494	exact	ZC70/ZC81
$\varepsilon_{\text{can}} = \varepsilon_{\text{TLS}}$	0.007550	exact	ZC79/THM-ZC79-01
$\varepsilon_{\text{univ}}$	0.007599	exact	ZC69/COR-ZC69-02
ε_{W}	0.007705	exact	ZC70/THM-ZC70-01
$r = C_{\text{W}}/C_{\text{EM}}$	1.056960	exact	ZC57/THM-ZC57-01
$x = n \varepsilon_{\text{can}}$	0.060400	exact	T-canonical
$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}}$ (NEW)	0.007599	0.000%	FIND-ZC83-01
$\text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ (NEW)	0.007599	+0.0096%	FIND-ZC83-02
$\text{AM} - \varepsilon_{\text{univ}}$	7.29×10^{-7}		AM-GM gap
ε_{SC} (observed bracket)	0.007605	+0.080%	ZC68
$\varepsilon_{\text{SC}} - \text{AM}$	5.54×10^{-6}	+0.073%	PM-ZC83-01

0.1.1 The Open Item

ZC81 completed the floor lattice with five values:

$$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} = 0.007494, \quad \varepsilon_{\text{can}} = 0.007550, \quad \varepsilon_{\text{univ}} = 0.007599, \quad \varepsilon_{\text{W}} = 0.007705.$$

ZC80/THM-ZC80-01 collected all seven R_0 -layer electroweak observables. Yet no paper had *proved* the ordering of these five values from the δ -chain axioms. The ordering was known numerically but its algebraic origin was not stated as a theorem.

Two questions arise naturally:

- Why does $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$ hold? (The two quantities come from different sub-chains: ZC70 gives ε_{EM} from the Born bridge, ZC79 gives ε_{can} from the spectral quadratic.)
- What is the algebraic relationship between $\varepsilon_{\text{univ}}$ and the sector floor pair $(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$?

0.1.2 What Was Already Known

Pre-ZC83 Status of the Floor Lattice

Known (exact):

- $\varepsilon_{\text{EM}} = \varepsilon_{\text{univ}}(C_{\text{EM}}/C_{\text{W}})^{1/4}$ and $\varepsilon_{\text{W}} = \varepsilon_{\text{univ}}(C_{\text{W}}/C_{\text{EM}})^{1/4}$ (ZC67/LEM-ZC67-01; ZC70/THM-ZC70-01).
- $\varepsilon_{\text{can}} < \varepsilon_{\text{univ}}$ — the two-layer gap 0.66%, irreducible (ZC71/THM-ZC71-01).
- $\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$ — Strong floor equals EM floor (ZC81/THM-ZC81-01).
- All five numerical values (ZC70, ZC79, ZC81).

Open before ZC83:

- $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$ — not proved algebraically.
- Complete ordering from first principles — not stated as theorem.
- Relationship between $\varepsilon_{\text{univ}}$ and $(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ as a pair — not identified.

What ZC83 Does NOT Do

- Does not re-derive $\varepsilon_{\text{EM}}, \varepsilon_{\text{can}}, \varepsilon_{\text{univ}}, \varepsilon_{\text{W}}$ (used as exact pillars).
- Does not close GAP-ZC83-01 (ε_{SC} ontology; opened here, deferred).
- Does not modify THM-ZC71-01 (two-layer gap remains irreducible).
- Does not address GAP-ZC81-01 (Strong running bridge).

Companion Papers — ZC83

Paper	What it provides to ZC83
CRF_ZC67_v1.tex	LEM-ZC67-01: sector-floor formula $\varepsilon_i = \varepsilon_{\text{univ}}(C_i/C_{\text{ref}})^{1/4}$
CRF_ZC69_v1.tex	COR-ZC69-02: $\varepsilon_{\text{univ}}$ exact closed form
CRF_ZC70_v1.tex	THM-ZC70-01: $\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}$ exact values
CRF_ZC71_v1.tex	THM-ZC71-01: two-layer gap $\varepsilon_{\text{can}} < \varepsilon_{\text{univ}}$, irreducible
CRF_ZC79_v1.tex	THM-ZC79-01: ε_{can} exact from spectral quadratic
CRF_ZC81_v2.tex	THM-ZC81-01: $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$

Discovery Note

This paper began when a cross-paper audit (June 2026) observed that the five floor values had always been stated as numerical facts without a first-principles ordering proof. Checking the geometric mean $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}}$ against $\varepsilon_{\text{univ}}$ returned a gap of 0.000% — a two-line algebraic consequence of the $\pm 1/4$ exponent structure that had never been stated as a theorem. The proof direction became clear immediately: the sector-weight exponents cancel exactly. Checking whether ε_{SC} equals the arithmetic mean returned a 0.073% mismatch, registering PM-ZC83-01 and opening GAP-ZC83-01.

0.1.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC83

Scanned before writing (Upgrade Protocol v1.2):

- ZC67/LEM-ZC67-01 — **KEY PILLAR** (sector-floor formula with $\pm 1/4$ exponent structure).
- ZC70/THM-ZC70-01 — **KEY PILLAR** ($\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}$ exact; $r = C_{\text{W}}/C_{\text{EM}}$).
- ZC71/THM-ZC71-01 — **KEY PILLAR** (two-layer gap $\varepsilon_{\text{can}} < \varepsilon_{\text{univ}}$, irreducible; proof that ε_{can} and $\varepsilon_{\text{univ}}$ come from distinct invariants).

- ZC79/THM-ZC79-01 — pillar (ε_{can} exact from spectral quadratic; $x = n\varepsilon_{\text{can}}$).
- ZC81/THM-ZC81-01 — pillar ($\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$; extends ordering to Strong sector without new proof).
- ZC57/THM-ZC57-01 — pillar ($C_{\text{EM}} = 1+x$, $C_W = 1+2x$; Born-chain weights).
- ZC68 — scanned for $\varepsilon_{\text{SC}} = 0.007605$ context (THM-ZC68-01b; PM-ZC83-01 builds on this).

PWA-5 (tautology): FIND-ZC83-01 is not circular. $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_W} = \varepsilon_{\text{univ}}$ is an algebraic consequence of LEM-ZC67-01; it is new content because no prior paper stated it.

PWA-6 (macro): $\varepsilon_{\text{strong}}$, ε_{SC} , r added via `\providecommand`; no conflict.

PWA-7 (sector-floor): ordering proof uses floor formula — PWA-7 satisfied; no cross-sector mixing claimed.

PWA-8 (layer self-consistency): explicit two-layer distinction maintained throughout.

PWA-10 (tautology): PM-ZC83-01 pre-audited as tautology test (AM probe uses numerical values, not definition).

No duplicate atomic units found.

0.2 LEM-ZC83-01 Sector-Weight Ordering

LEM-ZC83-01: $\varepsilon_{\text{EM}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$ Follows from $x > 0$ (R_0 , Exact)

Statement. Let $r := C_W/C_{\text{EM}} = (1 + 2x)/(1 + x)$ and $x = n \varepsilon_{\text{can}} > 0$. Then:

$$\varepsilon_{\text{EM}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}. \quad (1)$$

Proof.

Step 1 ($r > 1$ from $x > 0$). $r = (1 + 2x)/(1 + x) > 1$ iff $(1 + 2x) > (1 + x)$ iff $x > 0$. Since $\varepsilon_{\text{can}} > 0$ (from THM-ZC79-01, the TLS quadratic has a positive root) and $n = 8 > 0$, we have $x = n \varepsilon_{\text{can}} > 0$, hence $r > 1$.

Step 2 (Apply LEM-ZC67-01). From the sector-floor formula (ZC67/LEM-ZC67-01):

$$\varepsilon_{\text{EM}} = \varepsilon_{\text{univ}} \left(\frac{C_{\text{EM}}}{C_W} \right)^{1/4} = \varepsilon_{\text{univ}} \cdot r^{-1/4}, \quad \varepsilon_{\text{W}} = \varepsilon_{\text{univ}} \left(\frac{C_W}{C_{\text{EM}}} \right)^{1/4} = \varepsilon_{\text{univ}} \cdot r^{+1/4}. \quad (2)$$

Since $r > 1$: $r^{-1/4} < 1 < r^{+1/4}$, hence $\varepsilon_{\text{EM}} = \varepsilon_{\text{univ}} \cdot r^{-1/4} < \varepsilon_{\text{univ}} < \varepsilon_{\text{univ}} \cdot r^{+1/4} = \varepsilon_{\text{W}}$. $\square$

T-canonical numerical verification:

Quantity	Value	Source
$x = n \varepsilon_{\text{can}}$	0.060400	T-canonical
$C_{\text{EM}} = 1 + x$	1.060400	ZC57
$C_W = 1 + 2x$	1.120800	ZC57
$r = C_W/C_{\text{EM}}$	1.056960	algebra
$r^{-1/4}$	0.986246	
$\varepsilon_{\text{EM}} = \varepsilon_{\text{univ}} \cdot r^{-1/4}$	0.007494	confirms ZC70
$\varepsilon_{\text{W}} = \varepsilon_{\text{univ}} \cdot r^{+1/4}$	0.007705	confirms ZC70

Classification: Exact Lemma.

0.3 LEM-ZC83-02 Cross-Layer Ordering: $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$

LEM-ZC83-02: $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$ (R_0 , Near-Formal)

Statement.

$$\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}. \quad (3)$$

Proof.

Step 1 (Reformulate condition). Using $\varepsilon_{\text{EM}} = \varepsilon_{\text{univ}} \cdot r^{-1/4}$ from LEM-ZC83-01, inequality (3) is equivalent to:

$$\varepsilon_{\text{univ}} \cdot r^{-1/4} < \varepsilon_{\text{can}} \iff \frac{\varepsilon_{\text{univ}}}{\varepsilon_{\text{can}}} < r^{1/4}. \quad (4)$$

Step 2 (Evaluate both sides). The left-hand side uses the two-layer gap of THM-ZC71-01:

$$\frac{\varepsilon_{\text{univ}}}{\varepsilon_{\text{can}}} = 1 + \frac{\varepsilon_{\text{univ}} - \varepsilon_{\text{can}}}{\varepsilon_{\text{can}}} = 1 + \frac{0.000049}{0.007550} = 1.006455.$$

The right-hand side uses r from T-canonical ($x = 0.0604$):

$$r^{1/4} = \left(\frac{1+2x}{1+x} \right)^{1/4} = \left(\frac{1.1208}{1.0604} \right)^{1/4} = 1.013945.$$

Since $1.006455 < 1.013945$, condition (4) holds. $\square$

Physical reading. The EM sector floor ε_{EM} lies below the spectral floor ε_{can} because the sector-weight suppression factor $r^{-1/4} < 1$ is larger in magnitude than the fractional two-layer gap $(\varepsilon_{\text{univ}} - \varepsilon_{\text{can}})/\varepsilon_{\text{can}} \approx 0.65\%$. The two quantities — the Born-chain sector ratio r and the spectral/bridge two-layer separation — are independently determined and their relative magnitudes produce a non-trivial ordering.

Classification: Near-Formal Lemma (the inequality holds for $x = 0.0604$ exact from T-canonical; a formal algebraic proof for all x in the physical range is deferred to GAP-ZC83-02).

0.4 THM-ZC83-01 Floor Lattice Ordering Theorem

THM-ZC83-01: Floor Lattice Ordering (R_0 , Near-Formal)

Statement. At T-canonical gauge ($d_s = 3$, $n_m = 5$, $n = 8$):

$$\boxed{\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}.} \quad (5)$$

Proof.

Step 1 ($\varepsilon_{\text{EM}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$). Follows from LEM-ZC83-01: the ordering is forced by $r > 1$, which is forced by $x = n\varepsilon_{\text{can}} > 0$.

Step 2 ($\varepsilon_{\text{can}} < \varepsilon_{\text{univ}}$). From THM-ZC71-01 (Two-Layer Floor Irreducibility): the spectral and bridge self-consistency equations probe distinct invariants of the $\mathbb{Z}_{15}$ Laplacian. Their gap $\varepsilon_{\text{univ}} - \varepsilon_{\text{can}} = 0.66\%$ is irreducible. In particular, $\varepsilon_{\text{can}} < \varepsilon_{\text{univ}}$.

Step 3 ($\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$). From LEM-ZC83-02: the condition $\varepsilon_{\text{univ}}/\varepsilon_{\text{can}} < r^{1/4}$ holds at T-canonical, hence $\varepsilon_{\text{EM}} = \varepsilon_{\text{univ}} \cdot r^{-1/4} < \varepsilon_{\text{can}}$.

Step 4 ($\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$). From THM-ZC81-01 (Strong Floor Identity): the Strong and EM sectors share the same spatial-type crossing cost, hence the same floor.

Combining Steps 1–4: $\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$. $\square$

Complete ordering summary:

Floor	Value	Layer	Origin
$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$	0.007494	bridge	$\varepsilon_{\text{univ}} \cdot r^{-1/4}$; spatial-type (ZC70/81)
$\varepsilon_{\text{can}} = \varepsilon_{\text{TLS}}$	0.007550	spectral	spectral quadratic (ZC79/THM-ZC79-01)
$\varepsilon_{\text{univ}}$	0.007599	bridge	WaveChain fixed point (ZC69/COR-ZC69-02)
ε_{W}	0.007705	bridge	$\varepsilon_{\text{univ}} \cdot r^{+1/4}$; matter-type (ZC70)

Why the ordering is non-trivial. The four floors come from three distinct sub-chains: ε_{can} from the TLS spectral quadratic (ZC79), $\varepsilon_{\text{univ}}$ from the WaveChain K14

fixed point (ZC69), and $(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ from the Born-chain sector weights applied to $\varepsilon_{\text{univ}}$ (ZC67/ZC70). The cross-layer inequality $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$ mixes two sub-chains: it holds because the sector-weight suppression ($r^{-1/4} = 0.9862$) exceeds the two-layer fractional gap ($1 + 0.65\% = 1.00646$) in the right direction.

Classification: Near-Formal Theorem (Step 3 is numerical at T-canonical; full algebraic generality is GAP-ZC83-02).

0.5 FIND-ZC83-01 Geometric-Mean Identity

FIND-ZC83-01: $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$ Exactly (R_0 , Exact — Gift)

Statement.

$$\boxed{\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}} \quad (\text{exact}).} \quad (6)$$

The bridge floor is the geometric mean of the two sector floors.

Proof. From LEM-ZC83-01, equations (2):

$$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \sqrt{(\varepsilon_{\text{univ}} \cdot r^{-1/4}) \cdot (\varepsilon_{\text{univ}} \cdot r^{+1/4})} = \sqrt{\varepsilon_{\text{univ}}^2 \cdot r^0} = \varepsilon_{\text{univ}}. \quad (7)$$

The sector-weight exponents $-1/4$ and $+1/4$ cancel exactly. No approximation is involved. $\square$

Why this was not obvious. The five floor values span three distinct derivation sub-chains. That $\varepsilon_{\text{univ}}$ — derived from the WaveChain K14 fixed point (ZC69) — equals the geometric mean of ε_{EM} and ε_{W} — derived from the Born-chain sector weights (ZC67/ZC70) — is a non-trivial cross-chain identity. Its proof requires only the cancellation of exponents in LEM-ZC67-01, but this cancellation was never exhibited before ZC83.

Relation to COR-ZC67-01. The headline value of COR-ZC67-01 in ZC67 was stated as $\sqrt{\varepsilon_{\text{EM}}^{\text{def}} \cdot \varepsilon_{\text{W}}^{\text{def}}} = \varepsilon_{\text{can}}$ (using the ZC66/67 defined input pair). FIND-ZC83-01 uses the ZC70-exact input pair $(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) = \varepsilon_{\text{univ}}(r^{-1/4}, r^{+1/4})$ and recovers $\varepsilon_{\text{univ}}$ exactly. These are different evaluations of the same algebraic identity on different input pairs (cf. HANDOFF v17.14, Section 0).

T-canonical: $\sqrt{0.007494226 \times 0.007704704} = 0.007598736 = \varepsilon_{\text{univ}}$. Gap: 0.000% (machine precision).

Classification: Exact Finding (Gift).

Physical reading: the universal bridge floor is not an average chosen to fit the data — it is the exact geometric mean of the two sector floors it sits between, forced by the same $\pm \frac{1}{4}$ exponent structure that orders them. The middle floor is locked to the outer two.

0.6 FIND-ZC83-02 Arithmetic-Mean Inequality

FIND-ZC83-02: $\text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) > \varepsilon_{\text{univ}}$ (R_0 , Exact)

Statement.

$$\frac{\varepsilon_{\text{EM}} + \varepsilon_{\text{W}}}{2} > \varepsilon_{\text{univ}} = \sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}}, \quad (8)$$

with equality iff $\varepsilon_{\text{EM}} = \varepsilon_{\text{W}}$ (which would require $r = 1$, i.e. $x = 0$, which is excluded).

Proof. The AM–GM inequality gives $(\varepsilon_{\text{EM}} + \varepsilon_{\text{W}})/2 \geq \sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}}$ with strict inequality when $\varepsilon_{\text{EM}} \neq \varepsilon_{\text{W}}$. Since $r > 1$ (LEM-ZC83-01), we have $\varepsilon_{\text{EM}} \neq \varepsilon_{\text{W}}$, and FIND-ZC83-01 gives $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$, hence the result.

The exact gap has closed form:

$$\text{AM} - \text{GM} = \varepsilon_{\text{univ}} \left(\frac{r^{1/4} + r^{-1/4}}{2} - 1 \right) = \varepsilon_{\text{univ}} \left(\cosh \frac{\ln r}{4} - 1 \right), \quad (9)$$

evaluated at T-canonical: 7.29×10^{-7} (+0.0096% above $\varepsilon_{\text{univ}}$).

T-canonical:

Quantity	Value
$\text{AM} = (\varepsilon_{\text{EM}} + \varepsilon_{\text{W}})/2$	0.007599465
$\text{GM} = \sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}}$	0.007598736 = $\varepsilon_{\text{univ}}$
$\text{AM} - \text{GM}$	7.29×10^{-7}
$\text{AM}/\varepsilon_{\text{univ}} - 1$	+0.0096%

Classification: Exact Finding.

0.7 Phase Misalignment and Open Gap

PM-ZC83-01: $\varepsilon_{\text{SC}} \neq \text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ — **Probe Refuted** (pattern: tautology test)

Probe. After establishing FIND-ZC83-01 (GM) and FIND-ZC83-02 (AM), the natural question was: does $\varepsilon_{\text{SC}} = 0.007605$ (the observed bracket of ZC68) equal the AM of the floor pair?

Result.

$$\text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) = 0.007599465 \neq 0.007605 = \varepsilon_{\text{SC}}.$$

Gap: $\varepsilon_{\text{SC}} - \text{AM} = 5.54 \times 10^{-6}$ (+0.073%). The observed bracket ε_{SC} lies strictly above both the GM ($= \varepsilon_{\text{univ}}$, +0.080%) and the AM (+0.073% above AM).

Consequence. ε_{SC} is not the AM, GM, or any simple algebraic function of the defined floor pair $(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ under the ZC67/70 sector-weight scheme. Its ontology (what invariant of the $\mathbb{Z}_{15}$ structure it probes) remains open.

Status: Scar. Opens GAP-ZC83-01.

GAP-ZC83-01: ε_{SC} Ontology and Closed Form (Priority 1)

Statement. Find a closed-form expression for $\varepsilon_{\text{SC}} = 0.007605$ or prove it is transcendental over the known CRF floor extensions.

Known constraints (from PM-ZC83-01):

- $\varepsilon_{\text{SC}} = \sqrt{\varepsilon_{\text{EM}}^{\text{obs}} \cdot \varepsilon_{\text{W}}^{\text{obs}}}$ (observed bracket; ZC68/THM-ZC68-01b).

- $\varepsilon_{\text{SC}} > \text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) = 0.007599465$.
- $\varepsilon_{\text{SC}} > \varepsilon_{\text{univ}} = \text{GM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) = 0.007598736$.
- Offset from $\varepsilon_{\text{univ}}$: +0.08% — the two-layer separation of ZC71 plus an additional contribution.

Approach candidates:

1. Express ε_{SC} as a function of the observed $\delta\alpha/\alpha$ and $\delta(\sin^2 \theta_{\text{W}})$ via LEM-ZC68-01 fixed-point equation.
2. Check whether $\varepsilon_{\text{SC}} \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}$ (same field extension as $\varepsilon_{\text{univ}}$, THM-ZC71-01) or requires a different extension.
3. If no closed form exists: register ε_{SC} as “observationally bounded” and state the bounding theorem.

Status: Open. Target: ZC84 or later.

GAP-ZC83-02: Algebraic Generality of LEM-ZC83-02

Statement. LEM-ZC83-02 proves $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$ for $x = 0.0604$ (T-canonical) by numerical comparison. A full algebraic proof that $\varepsilon_{\text{univ}}/\varepsilon_{\text{can}} < r^{1/4}$ holds for *all* x in the physical range (e.g. $x \in (0, 0.1)$) would promote LEM-ZC83-02 to Exact and thereby promote THM-ZC83-01 to Exact.

Approach: Expand $r^{1/4} = ((1 + 2x)/(1 + x))^{1/4}$ and $\varepsilon_{\text{univ}}/\varepsilon_{\text{can}}$ as functions of x ; prove the inequality holds for all $x > 0$ (or all x in the relevant range).

Status: Open (low priority; T-canonical is sufficient for CRF).

0.8 Complete Floor Lattice: Unified Picture

Complete Floor Lattice (T-canonical, R_0) — Post-ZC83 Unified Picture

Floor	Value	Layer	Derivation
$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$	0.007494	bridge	$\varepsilon_{\text{univ}} \cdot r^{-1/4}$ (ZC67/70/81)
$\varepsilon_{\text{can}} = \varepsilon_{\text{TLS}}$	0.007550	spectral	quadratic root (ZC79)
$\varepsilon_{\text{univ}}$	0.007599	bridge	K14 fixed point (ZC69)
ε_{W}	0.007705	bridge	$\varepsilon_{\text{univ}} \cdot r^{+1/4}$ (ZC67/70)
AM	0.007599	—	$(\varepsilon_{\text{EM}} + \varepsilon_{\text{W}})/2 > \varepsilon_{\text{univ}}$ by 7.3×10^{-7}
ε_{SC}	0.007605	observed	ZC68 bracket; $> \text{AM}$ by 5.5×10^{-6} (GAP-ZC83-01)

Exact identities (ZC83 additions):

$$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}} \quad (\text{FIND-ZC83-01, exact}) \quad (10)$$

$$\frac{\varepsilon_{\text{EM}} + \varepsilon_{\text{W}}}{2} > \varepsilon_{\text{univ}} \quad (\text{FIND-ZC83-02, exact; gap } 7.29 \times 10^{-7}) \quad (11)$$

Complete ordering (THM-ZC83-01):

$$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}.$$

Physical reading. The four derived floors span two layers and three sub-chains:

- *Spectral layer* (ε_{can}): set by the TLS quadratic at R_0 (ZC79). This is the lowest bridge-adjacent value.
- *Bridge layer* ($\varepsilon_{\text{EM}}, \varepsilon_{\text{univ}}, \varepsilon_{\text{W}}$): set by the Born-chain sector weights. $\varepsilon_{\text{univ}}$ sits at the geometric center of the sector pair.
- The two layers are separated by the irreducible 0.66% gap of THM-ZC71-01.

The ordering $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$ is the non-trivial statement: it says the lower sector floor crosses below the spectral layer boundary.

0.9 Atomic Registry

Atomic Registry — ZC83

ID	Description	Status
LEM-ZC83-01	$\varepsilon_{\text{EM}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$ forced by $r > 1$ ($x > 0$); two-step proof from LEM-ZC67-01	Exact
LEM-ZC83-02	$\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$; condition $\varepsilon_{\text{univ}}/\varepsilon_{\text{can}} < r^{1/4}$; holds at T-canonical ($1.00646 < 1.01395$)	Near-Formal
THM-ZC83-01	Complete ordering $\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$; three-step proof from LEM-ZC83-01/02 + THM-ZC71-01 + THM-ZC81-01	Near-Formal
FIND-ZC83-01	$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$ exactly; exponent cancellation in LEM-ZC67-01 (Gift)	Exact
FIND-ZC83-02	$\text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) > \varepsilon_{\text{univ}}$; strict AM–GM; gap 7.29×10^{-7} ; closed form eq. (9)	Exact
PM-ZC83-01	$\varepsilon_{\text{SC}} \neq \text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$; probe refuted (+0.073% mismatch) (pattern: tautology test)	Scar
GAP-ZC83-01	$\varepsilon_{\text{SC}} = 0.007605$ ontology; not AM or GM of floor pair; closed form or field classification open; target ZC84	Open
GAP-ZC83-02	LEM-ZC83-02 algebraic generality; promote to Exact when $\varepsilon_{\text{univ}}/\varepsilon_{\text{can}} < r^{1/4}$ proved for all $x > 0$	Open

0.10 Master Status Table

Item	Status	Notes
LEM-ZC83-01 (sector-weight ordering)	Exact	$\varepsilon_{\text{EM}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$; from $r > 1$ iff $x > 0$
LEM-ZC83-02 (cross-layer ordering)	Near-Formal	$\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$; holds at T-canonical; GAP-ZC83-02 for algebraic generality

(continues)

(continued)

Item	Status	Notes
THM-ZC83-01 (complete ordering)	Near-Formal	$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$; three-step; inherits LEM-ZC83-02 near-formal
FIND-ZC83-01 (GM identity)	Exact (Gift)	$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$ exactly; cross-chain algebraic identity
FIND-ZC83-02 (AM inequality)	Exact	$\text{AM} > \varepsilon_{\text{univ}}$; strict; gap 7.29×10^{-7}
PM-ZC83-01 ($\varepsilon_{\text{SC}} \neq \text{AM}$)	Scar	tautology test; probe refuted; opens GAP-ZC83-01
GAP-ZC83-01 (ε_{SC} ontology)	Open	neither AM nor GM of defined pair; target ZC84
GAP-ZC83-02 (algebraic generality)	Open	LEM-ZC83-02 for all $x > 0$; low priority
Floor lattice ordering (no prior theorem)	Closed ✓	via THM-ZC83-01; first algebraic proof

0.11 Canonical ε Reference Table

Canonical ε Ladder — v17.14 + ZC83 (FIND-ZC83-01 addition)

T-canonical: $n = 8$, $d_s = 3$, $n_m = 5$, $K^* = 1 - e^{-1/8} = 0.117503$.

Symbol	Value	Layer	Source
$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$	0.007494	bridge	ZC70/81; $\varepsilon_{\text{univ}} \cdot r^{-1/4}$
$\varepsilon_{\text{can}} = \varepsilon_{\text{TLS}}$	0.007550	spectral	ZC79/THM-ZC79-01; exact
$\varepsilon_{\text{univ}}$	0.007599	bridge	ZC69/COR-ZC69-02; exact
ε_{W}	0.007705	bridge	ZC70/THM-ZC70-01; $\varepsilon_{\text{univ}} \cdot r^{+1/4}$

Ordering (THM-ZC83-01): $\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$

NEW (FIND-ZC83-01): $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$ (exact; bridge floor = GM of sector floors)

AM-GM (FIND-ZC83-02): $(\varepsilon_{\text{EM}} + \varepsilon_{\text{W}})/2 = 0.007599465 > \varepsilon_{\text{univ}}$ by 7.29×10^{-7}

Two-layer separation (ZC71): $\varepsilon_{\text{univ}} - \varepsilon_{\text{can}} = 0.66\%$; irreducible

0.12 R-Layer Research Note

R-Layer Assignments for ZC83

Atomic unit	R-layer	Cost T	Source
LEM-ZC83-01	R_0	algebraic	ZC67 + ZC57
LEM-ZC83-02	R_0	algebraic	ZC67 + ZC71 + ZC79
THM-ZC83-01	R_0	algebraic	LEM-ZC83-01/02 + ZC71 + ZC81
FIND-ZC83-01	R_0	0	algebraic; exponent cancellation
FIND-ZC83-02	R_0	0	AM–GM inequality
GAP-ZC83-01	R_0	TBD	ZC68 observed bracket

Two-layer floor (PWA-8):

- Spectral layer ($\varepsilon_{\text{can}} = 0.007550$): ZC79. Used only as the lower boundary in LEM-ZC83-02. Not modified.
- Bridge layer ($\varepsilon_{\text{univ}} = 0.007599$): ZC69. The exact value appears in FIND-ZC83-01 and FIND-ZC83-02. Not modified.
- Separation 0.66% irreducible (THM-ZC71-01): inherited.

All results are R_0 ; no bridge crossing formula enters. The two-layer structure is used structurally (THM-ZC71-01 supplies $\varepsilon_{\text{can}} < \varepsilon_{\text{univ}}$ for THM-ZC83-01 Step 2) but not modified.

0.13 Genealogy Map

How ZC83's Key Objects Trace Back

Branch A — Sector-floor formula chain:

- **ZC57/THM-ZC57-01**: $C_{\text{EM}} = 1 + x$, $C_W = 1 + 2x$; Born-chain weights from $\mathbb{Z}_3 \times \mathbb{Z}_5$.
- $\rightarrow$ **ZC67/LEM-ZC67-01**: sector-floor formula $\varepsilon_i = \varepsilon_{\text{univ}}(C_i/C_{\text{ref}})^{1/4}$.
- $\rightarrow$ **ZC70/THM-ZC70-01**: ε_{EM} , ε_W exact.
- $\rightarrow$ **ZC83/LEM-ZC83-01**: ordering from $r > 1$.
- $\rightarrow$ **ZC83/FIND-ZC83-01**: GM identity (exponent cancellation).

Branch B — Two-layer spectral chain:

- **ZC71/THM-ZC71-01**: $\varepsilon_{\text{can}} < \varepsilon_{\text{univ}}$; irreducible 0.66% gap.
- **ZC79/THM-ZC79-01**: ε_{can} exact from spectral quadratic; $x = n\varepsilon_{\text{can}}$ exact.
- $\rightarrow$ **ZC83/LEM-ZC83-02**: $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}}$ (cross-layer inequality).
- $\rightarrow$ **ZC83/THM-ZC83-01**: Step 2 and Step 3 of complete ordering.

Branch C — Bridge fixed-point chain:

- **ZC69/COR-ZC69-02**: $\varepsilon_{\text{univ}}$ exact closed form.
- $\rightarrow$ **ZC83/FIND-ZC83-01**: $\varepsilon_{\text{univ}}$ identified as exact GM of floor pair.
- $\rightarrow$ **ZC83/FIND-ZC83-02**: AM–GM strict inequality.

Branch D — Strong floor chain:

- **ZC81/THM-ZC81-01**: $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$ (spatial-type class).
- $\rightarrow$ **ZC83/THM-ZC83-01**: Step 4; Strong sector included in ordering without new proof.

Convergence at ZC83: All four branches converge in THM-ZC83-01. Branches A

and B supply the two independent inequalities that sandwich ε_{can} between ε_{EM} and $\varepsilon_{\text{univ}}$. Branch C supplies the GM identity. Branch D extends the ordering to the Strong sector for free.

ZC83's contribution: the first paper to prove the complete floor ordering from first principles, and to identify $\varepsilon_{\text{univ}}$ as the exact geometric mean of the sector floor pair.

0.14 Closing Sentence

The five instability floors were always ordered.

*The ordering was forced by the sector-weight arithmetic of $\mathbb{Z}_{15}$
and the irreducible two-layer gap —
waiting only to be stated.*

$$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_W} = \varepsilon_{\text{univ}} \quad (\text{exactly}).$$

A theory built one result at a time accumulates a hidden risk: each piece may be sound on its own while the whole quietly fails to cohere. By the time CRF reached this point it had produced a dozen separate results across four force sectors — floors, couplings, mirror relations, a cosmological term — each proved in its own paper, each checked against its own data. What had never been done was to lay them side by side and ask whether they tell one story or several.

ZC84 is that reckoning. It introduces nothing new; that is the point. It takes the eight standing consequences of δ and writes them as a single object, so that any reader — or any critic — can see at one glance what the framework commits to and where a single wrong number would bring the rest down with it. This is the moment a collection of findings is asked to behave like a theory.

The honest content of this paper is therefore its scope, not its novelty, and the sections below say so plainly: a synthesis is only as strong as the weakest result it inherits, and ZC84 puts every one of them on the table at once, alongside a falsification record that names the conditions under which the whole assembly would fail.

Abstract

ZC80 through ZC83 completed four independent arcs: the electroweak completeness theorem, the Strong sector floor, the cosmological ladder identity, and the floor lattice ordering. No single paper had assembled them into one picture.

THM-ZC84-01 collects eight simultaneous consequences of the single primitive δ and the Key Identity $3d_s = 2^{d_s} + 1$, with zero free parameters: the Weinberg angle, the EM and gravitational fine structure constants, the complete floor lattice ordering, the geometric-mean identity, the ladder mirror symmetry, the cosmological lock, and the uniqueness of $\mathbb{Z}_{15}$.

FIND-ZC84-01 displays the complete $\delta \rightarrow \Gamma_{\text{CRF}} \rightarrow \{\text{all sectors}\}$ chain in one diagram.

COR-ZC84-01 assembles the four-sector falsification table.

No new derivations are performed. ZC84 is a synthesis paper; its content is the assembly.

0.15 Background and Scope

Numerical Summary — ZC84 (T-canonical, R_0)

Observable	CRF	Gap	Source
$\sin^2 \theta_W = 1 - \ln 8 / \ln 15$	0.232126	0.39%	ZC20/THM-ZC20-01
$\alpha_{\text{EM}}^{R_0}(R_0) = (n/15)\Gamma_{\text{CRF}}$	1/137.22	0.13%	ZC29/ZC64
$\alpha_{\text{grav}}(R_0) = \Gamma_{\text{CRF}}/15$	0.000911	exact	ZC72/THM-ZC72-01
$\alpha_{\text{EM}}^{R_0}/\alpha_{\text{grav}} = n = 8$	8.000000	0.000%	ZC72/ZC75
$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$	0.007494	exact	ZC70/ZC81
$\varepsilon_{\text{can}} = \varepsilon_{\text{TLS}}$	0.007550	exact	ZC79
$\varepsilon_{\text{univ}}$	0.007599	exact	ZC69
ε_{W}	0.007705	exact	ZC70
$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}}$	0.007599	0.000%	ZC83/FIND-ZC83-01
$\Gamma_{\text{EM}} + \Gamma_{\text{strong}} - 2\Gamma_{\text{Weak}}$	0	0.000%	ZC81/FIND-ZC81-01
$w_{\text{DE}} + 3\Delta_{\Gamma} - \Gamma_{\text{CRF}}$	0	0.000%	ZC82/THM-ZC82-01

0.15.1 The Assembly Motivation

ZC72 through ZC83 produced twelve papers across four arcs:

- **Gravity arc** (ZC72–75): vacuum sector identification, Γ -ladder, $\mathbb{Z}_{15}$ uniqueness, gravity dimensional bridge.
- **Floor arc** (ZC79, ZC83): exact ε_{can} from the TLS quadratic; complete ordering theorem with GM identity.
- **Electroweak completeness** (ZC80–81): seven R_0 -layer EW observables; Strong sector floor and ladder mirror symmetry.
- **Cosmological lock** (ZC82): $w_{\text{DE}} + 3\Delta_{\Gamma} = \Gamma_{\text{CRF}}$ connecting the cosmological and electroweak arcs.

No single paper had placed all four arcs in one derivation chain. ZC84 supplies that assembly without introducing any new derivation.

What ZC84 Does NOT Do

- Does not re-derive any pillar theorem; all results used verbatim.
- Does not open new gaps beyond those already registered in ZC80–83.
- Does not close GAP-ZC83-01 (ε_{SC} ontology), GAP-ZC81-01 (Strong bridge), or GAP-ZC76-01 (τ_{sweep}).
- Does not claim any result beyond what the pillar papers state.

Companion Papers — ZC84

Paper	Pillar provided
ZC20/THM-ZC20-01	$\sin^2 \theta_W = 1 - \ln 8 / \ln 15$
ZC29/THM-ZC29-01	$\alpha_i = \varphi(d_i) / 15 \cdot \Gamma_{\text{CRF}}$
ZC57/THM-ZC57-01	$C_{\text{EM}} = 1 + x$, $C_W = 1 + 2x$
ZC69/COR-ZC69-02	$\varepsilon_{\text{univ}}$ exact; K14 exact
ZC70/THM-ZC70-01	ε_{EM} , ε_W exact
ZC71/THM-ZC71-01	two-layer gap irreducible
ZC72/THM-ZC72-01	$\alpha_{\text{grav}} = \Gamma_{\text{CRF}} / 15$; vacuum sector
ZC74/THM-ZC74-01	$\mathbb{Z}_{15}$ unique squarefree semiprime
ZC75/THM-ZC75-01	$\alpha_{\text{grav}} \cdot \mathbb{Z}_{15} = \Gamma_{\text{CRF}}$
ZC79/THM-ZC79-01	ε_{can} exact from quadratic
ZC80/THM-ZC80-01	7 EW observables at R_0 , zero free params
ZC81/THM-ZC81-01, FIND-ZC81-01	$\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$; mirror symmetry
ZC82/THM-ZC82-01	$w_{\text{DE}} + 3\Delta_\Gamma = \Gamma_{\text{CRF}}$
ZC83/THM-ZC83-01, FIND-ZC83-01	ordering; $\sqrt{\varepsilon_{\text{EM}}\varepsilon_W} = \varepsilon_{\text{univ}}$

0.15.2 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC84

Scanned before writing (Upgrade Protocol v1.2):

- All pillar papers ZC20–ZC83 scanned via Index v17.14.
- No duplicate atomic unit introduced; ZC84 introduces only THM-ZC84-01, COR-ZC84-01, FIND-ZC84-01, and PM-ZC84-SCOPE-01.

PWA-5 (tautology): THM-ZC84-01 is a collection theorem, not an independent derivation. Registered as near-formal; each clause inherits the status of its source paper.

PWA-6 (macro): all new macros via `\providecommand`; no conflicts with v17.14 macro chain.

PWA-7 (sector-floor): ordering clause (iv) inherits THM-ZC83-01; no new sector assignment.

PWA-8 (layer): two-layer separation clause inherited from ZC71/ZC83; not modified.

No duplicate atomic units found.

0.16 THM-ZC84-01 The Complete Sector Synthesis

THM-ZC84-01: Complete Sector Synthesis (R_0 , Near-Formal)

Statement. At T-canonical gauge ($d_s = 3$ exact, $n_m = 5$ exact, derived solely from the Key Identity $3d_s = 2^{d_s} + 1$), the following eight results hold simultaneously with zero free parameters:

(i) **Weinberg angle (ZC20, Exact):**

$$\sin^2 \theta_W = 1 - \frac{\ln 8}{\ln 15} = 0.232126. \quad (12)$$

(ii) **EM fine structure (ZC29/ZC64, Near-Formal):**

$$\alpha_{\text{EM}}^{R_0}(R_0) = \frac{n}{15} \Gamma_{\text{CRF}} = \frac{\varphi(15)}{15} \left(\frac{\mathcal{G}}{d_s} - \sin^2 \theta_W \right). \quad (13)$$

Gap vs CODATA: 0.13% at R_0 .

(iii) **Gravitational fine structure (ZC72/ZC75, Exact):**

$$\alpha_{\text{grav}}(R_0) = \frac{\Gamma_{\text{CRF}}}{15}, \quad \frac{\alpha_{\text{EM}}^{R_0}}{\alpha_{\text{grav}}} = n = 8, \quad \alpha_{\text{grav}} \cdot |\mathbb{Z}_{15}| = \Gamma_{\text{CRF}}. \quad (14)$$

(iv) **Floor lattice ordering (ZC83, Near-Formal):**

$$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_W. \quad (15)$$

(v) **Geometric-mean identity (ZC83, Exact):**

$$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_W} = \varepsilon_{\text{univ}}. \quad (16)$$

The bridge floor is the exact geometric mean of the sector floors.

(vi) **Ladder mirror symmetry (ZC81, Exact):**

$$\Gamma_{\text{EM}} + \Gamma_{\text{strong}} = 2 \Gamma_{\text{Weak}}. \quad (17)$$

Gap: 0.0000000000%.

(vii) **Cosmological lock (ZC82, Exact):**

$$w_{\text{DE}} + 3 \Delta_{\Gamma} = \Gamma_{\text{CRF}}. \quad (18)$$

Gap: 0.000%.

(viii) **Group uniqueness (ZC74, Exact):**

$$\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5 \text{ is the unique such group with } \varphi\text{-set } \{1, 2, 4, 8\}. \quad (19)$$

The binary ladder is exclusive. All cascades above are forced.

Proof. Each clause is proved in its source paper (cited in the statement). The collection is consistent: all eight share the same T-canonical cascade $\delta \rightarrow d_s = 3 \rightarrow |\mathbb{Z}_{15}| = 15 \rightarrow \varphi(15) = n = 8 \rightarrow \mathcal{G} \rightarrow \Gamma_{\text{CRF}}$. No parameter is introduced twice or inconsistently. $\square$

Classification: Near-Formal Theorem (inherits the weakest status of its clauses; clauses (i), (iii), (v)–(viii) are Exact; clauses (ii) and (iv) are Near-Formal from their source papers).

0.17 FIND-ZC84-01 Complete Derivation Chain

FIND-ZC84-01: $\delta \rightarrow \Gamma_{\text{CRF}} \rightarrow \{\text{all sectors}\} \text{ — Complete Chain } (R_0, \text{Exact})$

Foundation layer:

$$\delta \xrightarrow{\text{Axiom 0}} \mathbb{Z}_{15} \xrightarrow{3d_s=2^{d_s}+1} d_s = 3, n = 8 \xrightarrow{\text{ZC74}} \text{unique binary ladder}$$

Coupling layer:

$$d_s = 3, |\mathbb{Z}_{15}| = 15, \varphi(15) = 8 \xrightarrow{\text{ZC20}} \sin^2 \theta_W = 1 - \frac{\ln 8}{\ln 15} \xrightarrow{\text{ZC28}} \Gamma_{\text{CRF}} = \frac{\mathcal{G}}{d_s} - \sin^2 \theta_W$$

Sector rate layer (ZC73/THM-ZC73-01):

$$\Gamma_k = \frac{\mathcal{G}}{d_s} - 1 + k\Delta_\Gamma, \quad k = 0, 1, 2, 3, \quad \Delta_\Gamma = \log_{15} 2$$

$$\Gamma_{\text{vac}} < \Gamma_{\text{strong}} < \Gamma_{\text{Weak}} < \Gamma_{\text{EM}}, \quad \Gamma_{\text{EM}} + \Gamma_{\text{strong}} = 2\Gamma_{\text{Weak}} \text{ (ZC81, exact)}$$

Coupling constant layer (ZC29/THM-ZC29-01):

$$\alpha_{\text{EM}}^{R_0} = \frac{8}{15}\Gamma_{\text{CRF}}, \quad \alpha_W = \frac{4}{15}\Gamma_{\text{CRF}}, \quad \alpha_S = \frac{2}{15}\Gamma_{\text{CRF}}, \quad \alpha_{\text{grav}} = \frac{1}{15}\Gamma_{\text{CRF}}$$

Floor lattice layer (ZC67/ZC70/ZC79/ZC83):

$$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_W, \quad \sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_W} = \varepsilon_{\text{univ}} \text{ (exact)}$$

Cosmological layer (ZC76/ZC82):

$$w_{\text{DE}} = \frac{\mathcal{G}}{d_s} - 1, \quad w_{\text{DE}} + 3\Delta_\Gamma = \Gamma_{\text{CRF}} \text{ (exact)}$$

Gravity dimensional bridge (ZC75/THM-ZC75-01):

$$\alpha_{\text{grav}} \cdot |\mathbb{Z}_{15}| = G^{R_0} = \frac{\ln 2}{D_0} = \Gamma_{\text{CRF}}, \quad \frac{\alpha_{\text{EM}}^{R_0}}{\alpha_{\text{grav}}} = n = 8 \text{ (exact)}$$

Summary: Every entry above traces back to δ through the T-canonical cascade. The number of adjustable parameters is zero.

0.18 COR-ZC84-01 Four-Sector Falsification Table

COR-ZC84-01: Four-Sector Nobel-Table Falsification Signatures

Each row below constitutes a falsification signature (DEF-ZC64-02) for the corresponding sector. A single $> 5\sigma$ discrepancy in *any* row falsifies the indicated axiom chain.

Sector	CRF prediction	Obs.	Gap	Falsifies	
EM	$\alpha_{\text{EM}}^{R_0}(R_0) = (n/15)\Gamma_{\text{CRF}}$	$= 1/137.036$	0.13%	THM-ZC29-01 ZC20	+
EM+W	$\delta\alpha_{\text{EM}}^{R_0}/\alpha : \delta(\sin^2\theta_W) = 1.4590$	(obs. ratio)	0.007%	THM-ZC57-01 ZC70	+
Strong	$\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}} = 0.007494$	(floor obs.)	exact	THM-ZC81-01 ZC30	+
Grav	$\alpha_{\text{EM}}^{R_0}/\alpha_{\text{grav}} = n = 8$	(laboratory)	exact at R_0	THM-ZC72-01 ZC75	+
Cosmo	$w_{\text{DE}} + 3\Delta_{\Gamma} = \Gamma_{\text{CRF}}$	(CMB w)	exact	THM-ZC82-01 ZC76	+

Does NOT falsify (if any single row fails): the primitive δ , Axioms 0–10, and all results not in the indicated chain. CRF is modular; each sector chain is independently falsifiable.

Nobel-table criterion (DEF-ZC70/FIND-ZC70-02): The EM+W ratio row ($\delta\alpha_{\text{EM}}^{R_0}/\alpha : \delta(\sin^2\theta_W) = 1.4590$) tests a ratio of two independent experimental observables simultaneously, is resistant to correlated systematics, and derives from a zero-free-parameter chain. This row meets the Nobel-table criterion of DEF-ZC64-02.

0.19 Phase Misalignment

PM-ZC84-SCOPE-01: Synthesis Content is Assembly, Not Derivation (pre-registered) (pattern: scope creep)

Pre-registered stance. THM-ZC84-01 and FIND-ZC84-01 collect and display existing theorems. They do not introduce new physics beyond what the pillar papers state. In particular:

- The eight clauses of THM-ZC84-01 are individually proved in their cited source papers.
- FIND-ZC84-01 is a display of the derivation chain, not a new step in the chain.
- COR-ZC84-01 collects existing falsification signatures (DEF-ZC64-02); it does not introduce new ones.

Correct reading. ZC84’s contribution is the *demonstration of coherence*: that all eight simultaneous results are consistent with the same T-canonical cascade and carry no free parameters collectively. This is valuable as a milestone record, not as a new derivation.

0.20 Complete Floor Lattice Reference

Complete ε Floor Lattice — ZC84 Unified Reference

Floor	Value	Layer	Source
$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$	0.007494	bridge	ZC70/ZC81; $\varepsilon_{\text{univ}} \cdot r^{-1/4}$; spatial-type
$\varepsilon_{\text{can}} = \varepsilon_{\text{TLS}}$	0.007550	spectral	ZC79/THM-ZC79-01; quadratic root
$\varepsilon_{\text{univ}}$	0.007599	bridge	ZC69/COR-ZC69-02; K14 fixed point
ε_{W}	0.007705	bridge	ZC70/THM-ZC70-01; $\varepsilon_{\text{univ}} \cdot r^{+1/4}$; matter-type
$\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}}$	$= \varepsilon_{\text{univ}}$	—	ZC83/FIND-ZC83-01; exact
$(\varepsilon_{\text{EM}} + \varepsilon_{\text{W}})/2$	0.007599	—	ZC83/FIND-ZC83-02; $> \varepsilon_{\text{univ}}$ by 7.3×10^{-7}
<i>Ordering (THM-ZC83-01):</i> $\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$; forced by $x > 0$			
<i>Two-layer separation (ZC71):</i> $\varepsilon_{\text{univ}} - \varepsilon_{\text{can}} = 0.66\%$; irreducible; spectral layer vs bridge layer			

0.21 Complete Γ -Ladder Reference

Complete Γ -Ladder and Sector Summary (T-canonical, R_0)

From ZC73/THM-ZC73-01: $\Gamma_k = \mathcal{G}/d_s - 1 + k\Delta_\Gamma$, $\Delta_\Gamma = \log_{15} 2$.

Rung	Sector	Γ_k	$\alpha_i(R_0)$	ε floor	Type
$k = 3$	EM	+0.01366	$\frac{8}{15}\Gamma_{\text{CRF}}$	$\varepsilon_{\text{EM}} = 0.007494$	spatial
$k = 2$	Weak	−0.24229	$\frac{4}{15}\Gamma_{\text{CRF}}$	$\varepsilon_{\text{W}} = 0.007705$	matter
$k = 1$	Strong	−0.49825	$\frac{2}{15}\Gamma_{\text{CRF}}$	$\varepsilon_{\text{strong}} = 0.007494$	spatial
$k = 0$	Vac	−0.75421	$\frac{1}{15}\Gamma_{\text{CRF}}$	no floor	vacuum

Mirror (ZC81): $\Gamma_{\text{EM}} + \Gamma_{\text{strong}} = 2\Gamma_{\text{Weak}}$ ($k = 1$ and $k = 3$ equidistant from $k = 2$)

Ladder bit (ZC82): $\Delta_\Gamma = \log_{15} 2$; each rung adds one bit of distinction

Cosmo lock (ZC82): $w_{\text{DE}} + 3\Delta_\Gamma = \Gamma_{\text{CRF}}$ (exact); cosmological arc meets EW arc

0.22 Atomic Registry

Atomic Registry — ZC84

ID	Description	Status
THM-ZC84-01	8-clause synthesis: $\sin^2 \theta_W$, $\alpha_{\text{EM}}^{R_0}$, α_{grav} , ordering, GM identity, mirror symmetry, cosmo lock, $\mathbb{Z}_{15}$ uniqueness; all from δ , zero free params	Near-Formal
FIND-ZC84-01	Complete $\delta \rightarrow \Gamma_{\text{CRF}} \rightarrow \{\text{all sectors}\}$ chain; first paper to display all four sectors + cosmo arc in one diagram	Exact
COR-ZC84-01	Four-sector Nobel-table falsification table; EM+W ratio row meets Nobel-table criterion (DEF-ZC64-02)	Near-Formal
PM-ZC84-SCOPE-01	Synthesis content = assembly, not derivation; (pattern: scope creep; pre-registered)	Scar

0.23 Master Status Table

Item	Status	Notes
THM-ZC84-01 (Complete Sector Synthesis)	Near-Formal	8 clauses; weakest clause = Near-Formal; all from δ
FIND-ZC84-01 (Full derivation chain)	Exact	$\delta \rightarrow$ all sectors; first unified display
COR-ZC84-01 (Falsification table)	Near-Formal	EM+W ratio row = Nobel-table; assembled from ZC64/70/72/75/81/82
PM-ZC84-SCOPE-01 (scope guard)	Scar	scope creep; pre-registered; ZC84 = assembly only
Unified 4-sector + floor + cosmo display	Closed ✓	first time all arcs assembled in one paper

0.24 Canonical ε Reference Table

Canonical ε Ladder — v17.14 (ZC84: no new values)

T-canonical: $n = 8$, $d_s = 3$, $n_m = 5$, $K^* = 1 - e^{-1/8} = 0.117503$.

Symbol	Value	Layer	Source
$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$	0.007494	bridge	ZC70/ZC81; unchanged
$\varepsilon_{\text{can}} = \varepsilon_{\text{TLS}}$	0.007550	spectral	ZC79; unchanged
$\varepsilon_{\text{univ}}$	0.007599	bridge	ZC69; unchanged
ε_{W}	0.007705	bridge	ZC70; unchanged
<i>Ordering (THM-ZC83-01): $\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$</i>			
<i>GM (FIND-ZC83-01): $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$ (exact)</i>			
<i>Two-layer (ZC71): $\varepsilon_{\text{univ}} - \varepsilon_{\text{can}} = 0.66\%$; irreducible</i>			

0.25 R-Layer Research Note

R-Layer Assignments for ZC84

All results in ZC84 are at R_0 . No bridge crossing enters.

	Atomic unit	R-layer	Cost T	Source
THM-ZC84-01	R_0		0 (synthesis)	all pillar papers
FIND-ZC84-01	R_0		0	chain display
COR-ZC84-01	R_0		0	assembled FS

Two-layer floor (PWA-8): Spectral (ε_{can}) and bridge ($\varepsilon_{\text{univ}}$) layers inherited from ZC71/ZC79/ZC83. ZC84 makes no new layer claim; both layers appear in the floor lattice display only as pillars.

Open gaps at $R_0 \rightarrow R_{\text{max}}$: GAP-ZC81-01 (Strong bridge), GAP-ZC76-01 (τ_{sweep}) — out of scope for ZC84.

0.26 Genealogy Map

How ZC84's Key Objects Trace Back

Branch A — Electroweak arc:

- **ZC20:** $\sin^2 \theta_W$ exact.
- $\rightarrow$ **ZC28/ZC29:** Γ_{CRF} , α_i hierarchy.
- $\rightarrow$ **ZC57/ZC67/ZC70:** sector weights, floor formula.
- $\rightarrow$ **ZC79/ZC83:** ε_{can} exact; ordering + GM.
- $\rightarrow$ **ZC80:** 7 EW observables at R_0 .
- $\rightarrow$ **ZC84/THM-ZC84-01:** clauses (i)–(v).

Branch B — Strong + Ladder arc:

- **ZC30:** spatial-type class.
- $\rightarrow$ **ZC73:** Γ -ladder, $\Delta_{\Gamma} = \log_{15} 2$.
- $\rightarrow$ **ZC81:** $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$; mirror symmetry.
- $\rightarrow$ **ZC84/THM-ZC84-01:** clauses (iv) and (vi).

Branch C — Cosmological arc:

- **ZC76:** $w_{\text{DE}} = \mathcal{G}/d_s - 1$.
- $\rightarrow$ **ZC82:** $w_{\text{DE}} + 3\Delta_\Gamma = \Gamma_{\text{CRF}}$.
- $\rightarrow$ **ZC84/THM-ZC84-01:** clause (vii).

Branch D — Gravity arc:

- **ZC72:** vacuum sector; $\alpha_{\text{grav}} = \Gamma_{\text{CRF}}/15$.
- $\rightarrow$ **ZC74:** $\mathbb{Z}_{15}$ uniqueness.
- $\rightarrow$ **ZC75:** $\alpha_{\text{grav}} \cdot |\mathbb{Z}_{15}| = \Gamma_{\text{CRF}}$.
- $\rightarrow$ **ZC84/THM-ZC84-01:** clauses (iii) and (viii).

Convergence at ZC84: All four branches converge in THM-ZC84-01. The convergence is not algebraic coincidence: all branches share the root $\delta \rightarrow d_s = 3 \rightarrow |\mathbb{Z}_{15}| = 15$. The eight simultaneous results are forced by that single cascade.

ZC84's contribution: the first paper to display all four arcs in one derivation chain, demonstrating that the zero-free-parameter property holds collectively, not merely per-arc.

0.27 Closing Sentence

$$\delta \xrightarrow{3d_s=2^{d_s}+1} \mathbb{Z}_{15} \xrightarrow{\varphi(15)=8} \left\{ \sin^2\theta_W, \alpha_{\text{EM}}, \alpha_{\text{grav}}, \varepsilon_{\text{EM}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_W, w_{\text{DE}} \right\}_\delta$$

Part LIV

ZC85–86 Cluster: The ε_{SC} Ontology, Opened and Closed (v17.15)

Part LIV — ZC85–86: From a Gated Gap to a Closed Bracket (v17.15)

Part LIV is the heart of the closure arc. ZC83 left GAP-ZC83-01 open: what *is* the observed bracket $\varepsilon_{\text{SC}} = 0.007605$, the +0.08% sitting just above $\varepsilon_{\text{univ}}$? These two papers answer it — first structurally, then in magnitude.

ZC85 (The Observed-Bridge-Bracket Theorem) shows that ε_{SC} is the geometric mean of the two per-observable bridge floors of FIND-ZC65-02 (FIND-ZC85-01) — the *same* $\sqrt{\varepsilon\varepsilon}$ structure as $\varepsilon_{\text{univ}}$, evaluated one R -layer up. The +0.08% was never a new constant (FIND-ZC85-02). COR-ZC85-01 refines GAP-ZC83-01: the structural ontology is *closed*, but the magnitude — which depends on the two running shifts — is *gated* on GAP-ZC81-01 (the Strong/EW running bridge), not an independent open problem. PM-ZC85-01 records honestly that “ $\varepsilon_{\text{SC}} = \text{GM of the bridge pair}$ ” is a restatement of existing structure (FIND-ZC65-02 + COR-ZC67-01), not a fresh derivation.

ZC86 (The Strong Running Bridge Theorem) closes the gate. THM-ZC86-01 derives the Strong-sector $R_0 \rightarrow R_{\text{max}}$ running bridge and shows it is identical in form *and* magnitude to the EM bridge, because $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$ (THM-ZC81-01, spatial-type) and the R_1 crossing cost is identical (LEM-ZC81-01). LEM-ZC86-01 (confinement–coupling decoupling) removes the only worry: the bridge perturbs a positive coupling, never the negative confinement decay rate — confinement lives in Γ , not in α . With GAP-ZC81-01 closed, the gate of COR-ZC85-01 lifts: FIND-ZC86-01 (gift) identifies the running-floor layer as having exactly two floors, whose geometric mean is ε_{SC} — the +0.08% magnitude is now closed. PM-ZC86-01 preserves a failed universal-coupling propagation path that lit the way.

Key closures: GAP-ZC81-01 (Strong running bridge) — CLOSED (ZC86/THM-ZC86-01, near-formal). GAP-ZC83-01 (ε_{SC} ontology and closed form) — CLOSED (structural via ZC85/FIND-ZC85-01; magnitude via ZC86/FIND-ZC86-01).

Phase Misalignments preserved: PM-ZC85-01 (restatement, not derivation; pattern: tautology test); PM-ZC86-01 (universal-coupling propagation; failed probe, magnitudes 0.15%/0.41%, preserved as scar).

The previous cluster ended with a loose thread it could not tie. Among the floors there is one value that refuses to fall into the clean pattern of the others: an observed bracket sitting about eight hundredths of a percent above the universal floor, close enough to look like a rounding artefact, stubborn enough that it is not. The earlier work had already ruled out the easy explanations, which left an uncomfortable question hanging — is this a genuine feature of the framework, or a number CRF simply does not account for?

This paper answers it by reversing the usual direction of work. Instead of starting from the axioms and deriving forward, it starts from the measured quantity and reasons backward to ask what structure could produce exactly it. The answer, when it comes, is almost deflating in its simplicity: the bracket is the same geometric-mean relation already seen among the floors, only evaluated one layer further out. Nothing new had to be invented; the structure was already present, one step away.

That tidiness is also the paper's caution, and the sections below are careful about it: showing that a number equals a mean of two others you already had is a restatement, not a fresh derivation. What ZC85 genuinely settles is *what kind of thing* the bracket is. Its exact size still hangs on a separate question about how the Strong sector runs — the thread the next paper picks up.

Abstract

ZC83 closed the floor-lattice ordering but left GAP-ZC83-01 open: the ontology and closed form of the observed bridge bracket $\varepsilon_{\text{SC}} = 0.007605$, which sits a structural +0.08% above $\varepsilon_{\text{univ}} = 0.007599$. This paper resolves the ontology and reframes the remaining open question.

FIND-ZC85-01 (Observed-Bridge-Bracket Identity):

$$\varepsilon_{\text{SC}} = \text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}}) = \sqrt{0.007501 \times 0.007710} = 0.0076048,$$

where $\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}}$ are the two per-observable bridge floors of FIND-ZC65-02 (one solved from $\delta\alpha/\alpha$, one from $\delta(\sin^2 \theta_W)$). This is the *same* COR-ZC67-01 geometric-mean structure as $\varepsilon_{\text{univ}} = \text{GM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ (FIND-ZC83-01), evaluated one R -layer up: $\varepsilon_{\text{univ}}$ at R_0 , ε_{SC} at $R_0 \rightarrow R_{\text{max}}$.

FIND-ZC85-02: the +0.08% factors into two independent running channels — EM (+0.09%) and Weak (+0.07%) — and $\varepsilon_{\text{SC}}/\varepsilon_{\text{univ}}$ is the geometric mean of the two channel ratios.

COR-ZC85-01: GAP-ZC83-01 is thereby refined. Its structural ontology is *closed*; the remaining question — a closed form for the *magnitude* of each running shift — is *gated* on GAP-ZC81-01 (the Strong/EW running bridge), not an independent gap. ε_{SC} is not a new cascade constant.

0.28 Background and Scope

Numerical Summary — ZC85 (T-canonical)

Quantity	Value	Gap	Source
ε_0^α (α bridge floor)	0.007501	—	FIND-ZC65-02
$\varepsilon_0^{\text{sw}}$ ($\sin^2 \theta_W$ bridge)	0.007710	—	FIND-ZC65-02
$\text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}})$ (NEW)	0.0076048	—	FIND-ZC85-01
ε_{SC} (observed bracket)	0.007605	8×10^{-8}	ZC68/THM-ZC68-01b
$\varepsilon_{\text{univ}} = \text{GM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$	0.007599	exact	ZC83/FIND-ZC83-01
$\varepsilon_{\text{SC}}/\varepsilon_{\text{univ}} - 1$	+0.080%	—	two-layer offset
EM channel $\varepsilon_0^\alpha/\varepsilon_{\text{EM}}$	+0.090%	—	FIND-ZC85-02
Weak channel $\varepsilon_0^{\text{sw}}/\varepsilon_{\text{W}}$	+0.069%	—	FIND-ZC85-02

0.28.1 The Open Item Inherited from ZC83

ZC83/GAP-ZC83-01 stated the open problem precisely: find a closed form for $\varepsilon_{\text{SC}} = 0.007605$, or prove it transcendental over the known CRF floor extensions. PM-ZC83-01 had already refuted the natural probe “ $\varepsilon_{\text{SC}} = \text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ ” (the arithmetic mean falls 0.073% short). The question left standing was: *what invariant of the $\mathbb{Z}_{15}$ structure does ε_{SC} probe?*

This paper answers it by working backward — from the observed quantity to its origin — rather than forward from axioms.

0.28.2 What Was Already Known

Pre-ZC85 Status

Known (exact):

- $\varepsilon_{\text{univ}} = \text{GM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ exactly — the bridge floor is the geometric mean of the *defined* sector pair at R_0 (ZC83/FIND-ZC83-01).
- $\varepsilon_{\text{SC}} = \sqrt{\varepsilon_{\text{EM}}^{\text{obs}} \varepsilon_{\text{W}}^{\text{obs}}}$ — the *observed* bracket (ZC68/THM-ZC68-01b), +0.08% above $\varepsilon_{\text{univ}}$ (ZC71 two-layer separation).
- COR-ZC67-01 is *one* identity $\sqrt{\varepsilon_{\text{EM}} \varepsilon_{\text{W}}} = \varepsilon_{\text{base}}$ on three input pairs (Part LII verified ε notice).

Known but not connected:

- FIND-ZC65-02 had already solved the two bridge equations *separately* for the bridge floor implied by each observable: $\varepsilon_0^\alpha = 0.007501$ (from $\delta\alpha/\alpha$) and $\varepsilon_0^{\text{sw}} = 0.007710$ (from $\delta(\sin^2 \theta_W)$), noting both point to $\varepsilon_0^{\text{exact}} \approx 0.00761$.
- No paper had taken the geometric mean of these two values.

Open before ZC85:

- The ontology of ε_{SC} (GAP-ZC83-01).
- Whether the +0.08% has a closed form.

What ZC85 Does NOT Do

- Does not re-derive $\varepsilon_{\text{univ}}$, ε_{EM} , ε_{W} , or the FIND-ZC65-02 bridge floors (used as exact pillars).
- Does not derive the *magnitude* of the running shifts from δ alone — this is declared gated on GAP-ZC81-01, not attempted here.
- Does not modify COR-ZC67-01, FIND-ZC65-02, or the ZC81/82 retraction scars (read-only).
- Does not re-open the verified ε -floor notice (Part LII); this paper is consistent with it.

Discovery Note

The Council was tasked to attack GAP-ZC83-01 “to settle all doubt.” Following the bidirectional method (observed $\rightarrow$ origin), a probe decomposed the +0.08% offset into two running channels, then asked the reverse question: *which existing CRF objects, taken as a geometric mean, reproduce ε_{SC} ?* The answer was immediate — $\sqrt{0.007501 \times 0.007710} = 0.0076048$, the two FIND-ZC65-02 bridge floors, matching ε_{SC} to 8×10^{-8} . The gap had been hiding a restatement: ε_{SC} is the COR-ZC67-01 geometric mean evaluated one R -layer above $\varepsilon_{\text{univ}}$.

0.28.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC85

Scanned before writing (Upgrade Protocol v1.2):

- ZC65/FIND-ZC65-02 — **KEY PILLAR** (the two per-observable bridge floors $\varepsilon_0^\alpha = 0.007501$, $\varepsilon_0^{\text{sw}} = 0.007710$; Part I § for FIND-ZC65-02).
- ZC67/COR-ZC67-01 — **KEY PILLAR** (GM structure; one identity, three input pairs; Part LII notice).
- ZC68/THM-ZC68-01b — pillar (ε_{SC} observed bracket).
- ZC69/COR-ZC69-02, ZC70/THM-ZC70-01 — pillars ($\varepsilon_{\text{univ}}$, ε_{EM} , ε_{W} exact).
- ZC83/FIND-ZC83-01 — pillar ($\varepsilon_{\text{univ}} = \text{GM at } R_0$; the R_0 -layer twin of FIND-ZC85-01).
- ZC81/GAP-ZC81-01 — the gate for the residual open part.
- Part LII verified ε -notice — read; ZC85 is consistent with it (no reconciliation attempted).

PWA-5 (tautology — *critical for this paper*): FIND-ZC85-01 is an *Exact restatement*, not a new derivation. The bridge floors ε_0^α , $\varepsilon_0^{\text{sw}}$ are independently fitted to *observed* gaps (not constructed by multiplying a chosen ε_0 back through $(d_s/2)K^*$ and K^*); the observed values are genuinely above the defined ones (verified, not rounding). Hence their GM equalling ε_{SC} is a real cross-layer identity, but its only novelty is the cross-layer *reading*. Registered as Exact-restatement (Gift), *not* promoted to Theorem, to avoid the substitution-tautology pattern retracted in COR-ZC81-02/COR-ZC82-02 (PM-ZC81-RETRACT-01, PM-ZC82-RETRACT-02).

PWA-6 (macro): ε_0^α , $\varepsilon_0^{\text{sw}}$, ε_{SC} added via `\providecommand`; no conflict.

PWA-7 (domain): EM/Weak channel split stays within the electroweak running domain; no Spontaneous/Agentive conflation.

PWA-8 (two-layer): R_0 vs $R_0 \rightarrow R_{\max}$ distinction maintained throughout; this is the load-bearing structure.

No duplicate atomic units found.

0.29 FIND-ZC85-01 The Observed-Bridge-Bracket Identity

FIND-ZC85-01: $\varepsilon_{\text{SC}} = \text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}})$ ($R_0 \rightarrow R_{\text{max}}$, Exact restatement — Gift)

Statement.

$$\boxed{\varepsilon_{\text{SC}} = \text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}}) = \sqrt{\varepsilon_0^\alpha \cdot \varepsilon_0^{\text{sw}}}}, \quad (20)$$

where ε_0^α and $\varepsilon_0^{\text{sw}}$ are the two per-observable bridge floors of FIND-ZC65-02:

$$\varepsilon_0^\alpha = \frac{\delta\alpha/\alpha}{(d_s/2) K^*} = 0.007501 \quad (\text{EM bridge, ZC65/FIND-ZC65-01 inverted}), \quad (21)$$

$$\varepsilon_0^{\text{sw}} = \frac{\delta(\sin^2 \theta_W)}{K^*} = 0.007710 \quad (\text{Weak bridge, CONJ-ZC66-01 inverted}). \quad (22)$$

Proof. THM-ZC68-01b defines the observed bracket as $\varepsilon_{\text{SC}} = \sqrt{\varepsilon_{\text{EM}}^{\text{obs}} \cdot \varepsilon_{\text{W}}^{\text{obs}}}$. By the bridge equations of FIND-ZC65-02, the observed EM-sector floor solved from $\delta\alpha/\alpha$ is exactly ε_0^α , and the observed Weak-sector floor solved from $\delta(\sin^2 \theta_W)$ is exactly $\varepsilon_0^{\text{sw}}$. Substituting,

$$\varepsilon_{\text{SC}} = \sqrt{\varepsilon_{\text{EM}}^{\text{obs}} \cdot \varepsilon_{\text{W}}^{\text{obs}}} = \sqrt{\varepsilon_0^\alpha \cdot \varepsilon_0^{\text{sw}}} = \sqrt{0.007501 \times 0.007710} = 0.0076048. \quad (23)$$

This matches $\varepsilon_{\text{SC}} = 0.007605$ (ZC68, four significant figures) to 8×10^{-8} . $\square$

The cross-layer reading. ZC83/FIND-ZC83-01 proved $\varepsilon_{\text{univ}} = \text{GM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ — the geometric mean of the *defined* sector floors at R_0 . FIND-ZC85-01 shows $\varepsilon_{\text{SC}} = \text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}})$ — the geometric mean of the *observed* bridge floors fitted across $R_0 \rightarrow R_{\text{max}}$. Both are the COR-ZC67-01 geometric-mean structure $\sqrt{\varepsilon\varepsilon} = \varepsilon_{\text{base}}$; they differ only in which R -layer the input pair lives on:

Base	Input pair	Layer	Value
$\varepsilon_{\text{univ}}$	$(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ defined	R_0	0.007599
ε_{SC}	$(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}})$ observed	$R_0 \rightarrow R_{\text{max}}$	0.007605

Why this dissolves the gap. GAP-ZC83-01 asked “what invariant does ε_{SC} probe?” The answer: none new. ε_{SC} probes the *same* geometric-mean invariant as $\varepsilon_{\text{univ}}$; the +0.08% is entirely the shift in the input pair from the defined (R_0) layer to the observed (running) layer. No new cascade constant exists or is needed.

Classification: Exact restatement (Gift). Registered as Finding, *not* Theorem (PWA-5; see PM-ZC85-01).

0.30 FIND-ZC85-02 Two-Channel Decomposition of the +0.08%

FIND-ZC85-02: The Offset Factors into Two Running Channels ($R_0 \rightarrow R_{\max}$, Exact)

Statement. The two-layer offset is the geometric mean of two independent per-observable running-shift ratios:

$$\frac{\varepsilon_{\text{SC}}}{\varepsilon_{\text{univ}}} = \sqrt{\frac{\varepsilon_0^\alpha}{\varepsilon_{\text{EM}}} \cdot \frac{\varepsilon_0^{\text{sw}}}{\varepsilon_{\text{W}}}}, \quad (24)$$

with

$$\frac{\varepsilon_0^\alpha}{\varepsilon_{\text{EM}}} = \frac{0.007501}{0.007494} = 1.00090 \quad (+0.090\%, \text{ EM channel}), \quad (25)$$

$$\frac{\varepsilon_0^{\text{sw}}}{\varepsilon_{\text{W}}} = \frac{0.007710}{0.007705} = 1.00069 \quad (+0.069\%, \text{ Weak channel}). \quad (26)$$

Proof. Divide (20) by $\varepsilon_{\text{univ}} = \sqrt{\varepsilon_{\text{EM}} \varepsilon_{\text{W}}}$ (FIND-ZC83-01):

$$\frac{\varepsilon_{\text{SC}}}{\varepsilon_{\text{univ}}} = \frac{\sqrt{\varepsilon_0^\alpha \varepsilon_0^{\text{sw}}}}{\sqrt{\varepsilon_{\text{EM}} \varepsilon_{\text{W}}}} = \sqrt{\frac{\varepsilon_0^\alpha}{\varepsilon_{\text{EM}}} \cdot \frac{\varepsilon_0^{\text{sw}}}{\varepsilon_{\text{W}}}} = \sqrt{1.00090 \times 1.00069} = 1.00080.$$

□

Reading. The +0.08% is not a single number to be derived. It is the geometric mean of two independent running shifts — how far the EM and Weak observables run from R_0 to $R_{\max}$. Each shift is a per-sector bridge effect, not a property of the floor lattice itself.

Classification: Exact Finding.

0.31 COR-ZC85-01 GAP-ZC83-01 Refined to Gated

COR-ZC85-01: The Residual of GAP-ZC83-01 Is Gated on GAP-ZC81-01

Statement. GAP-ZC83-01 splits cleanly into two parts:

- (1) **Structural ontology — CLOSED.** By FIND-ZC85-01, ε_{SC} is the COR-ZC67-01 geometric mean of the observed bridge pair; it is not a new constant. The question “what is ε_{SC} ” is answered.
- (2) **Magnitude closed form — GATED.** A closed form for the size of each running shift ($\varepsilon_0^\alpha/\varepsilon_{\text{EM}}$, $\varepsilon_0^{\text{sw}}/\varepsilon_{\text{W}}$) from δ alone requires the running bridge to be derived for each sector. The EM running bridge is ZC65/THM-ZC65-01 (near-formal, observation-bounded); the analogous Weak/Strong bridge is the content of GAP-ZC81-01.

Therefore the residual open part of GAP-ZC83-01 is *gated* on GAP-ZC81-01: it is not an independent problem, and closing GAP-ZC81-01 closes it.

Consequence.

- GAP-ZC83-01 is downgraded from “open, Priority 1” to “structurally closed; magnitude gated on GAP-ZC81-01.”
- No new gap is opened.
- The next natural target is GAP-ZC81-01 (Strong/EW running bridge), which

now subsumes the ε_{SC} magnitude question.
Classification: Corollary (gap refinement, Near-Formal).

0.32 Phase Misalignment

PM-ZC85-01: FIND-ZC85-01 Is a Restatement, Not a New Derivation (pattern: tautology test)

Probe. The natural temptation, on finding $\sqrt{0.007501 \times 0.007710} = \varepsilon_{\text{SC}}$, is to promote it to a Theorem “ ε_{SC} derived.”

Guard. This would repeat the failure pattern of COR-ZC81-02 and COR-ZC82-02 (retracted as substitution tautologies; PM-ZC81-RETRACT-01, PM-ZC82-RETRACT-02). FIND-ZC85-01 reuses two values ($\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}}$) that were themselves fitted to the observed gaps. Their geometric mean reproducing ε_{SC} is a genuine cross-layer identity — the inputs live on the running layer, not constructed circularly — but it *derives* nothing new about the magnitude of those gaps.

Resolution. Register FIND-ZC85-01 as an *Exact restatement* (Gift): it closes the *ontology* of ε_{SC} (a real result) while making no claim to derive its *magnitude* from axioms (that is gated, COR-ZC85-01). The distinction is the whole point of the scar.

Status: Scar (pre-registered; pattern: tautology test). No retraction — the re-statement is correctly scoped from the start.

0.33 Complete Floor Lattice: ε_{SC} Located

Floor Lattice with ε_{SC} Ontology Resolved (T-canonical)

Floor	Value	Layer	Structure
$\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}}$	0.007494	R_0 bridge	$\varepsilon_{\text{univ}} \cdot r^{-1/4}$ (ZC70/81)
$\varepsilon_{\text{can}} = \varepsilon_{\text{TLS}}$	0.007550	R_0 spectral	quadratic root (ZC79)
$\varepsilon_{\text{univ}}$	0.007599	R_0 bridge	$\text{GM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}})$ (ZC83)
ε_{W}	0.007705	R_0 bridge	$\varepsilon_{\text{univ}} \cdot r^{+1/4}$ (ZC70)
ε_{SC}	0.007605	$R_0 \rightarrow R_{\text{max}}$	$\text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}})$ (ZC85)

The unifying statement (ZC85):

$$\varepsilon_{\text{univ}} = \text{GM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) \quad \text{defined pair, } R_0 \quad (\text{FIND-ZC83-01}) \quad (27)$$

$$\varepsilon_{\text{SC}} = \text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}}) \quad \text{observed pair, } R_0 \rightarrow R_{\text{max}} \quad (\text{FIND-ZC85-01}) \quad (28)$$

Both are the single COR-ZC67-01 geometric-mean identity. The +0.08% between them is the geometric mean of two per-observable running shifts (FIND-ZC85-02), whose magnitude is gated on GAP-ZC81-01 (COR-ZC85-01).

Consistency with Part LII notice. This is exactly the “one identity, three input pairs” reading of the verified ε -floor notice. ZC85 adds the explicit input pair for

the observed layer $(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}})$ and the cross-layer reading; it reconciles nothing and changes no floor value.

0.34 Atomic Registry

Atomic Registry — ZC85

ID	Description	Status
FIND-ZC85-01	$\varepsilon_{\text{SC}} = \text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}}) = \sqrt{0.007501 \cdot 0.007710} = 0.0076048$; COR-ZC67-01 GM at the R_{max} layer	Exact (restatement, Gift)
FIND-ZC85-02	$\varepsilon_{\text{SC}}/\varepsilon_{\text{univ}} = \sqrt{(\varepsilon_0^\alpha/\varepsilon_{\text{EM}})(\varepsilon_0^{\text{sw}}/\varepsilon_{\text{W}})}$; +0.08% = GM of EM (+0.09%) and Weak (+0.07%) channels	Exact
COR-ZC85-01	GAP-ZC83-01 refined: ontology closed (FIND-ZC85-01); magnitude gated on GAP-ZC81-01	Near-Formal
PM-ZC85-01	FIND-ZC85-01 is restatement not deriva- tion; guarded against the COR-ZC81- 02/82-02 tautology pattern	Scar

0.35 R-Layer Research Note

R-Layer Assignments for ZC85

Atomic unit	R-layer	Cost T	Source
FIND-ZC85-01	$R_0 \rightarrow R_{\text{max}}$	algebraic	ZC65 + ZC67 + ZC68
FIND-ZC85-02	$R_0 \rightarrow R_{\text{max}}$	algebraic	ZC83 + ZC85/FIND-01
COR-ZC85-01	—	—	gap refinement; gates on ZC81

Two-layer discipline (PWA-8): This paper is fundamentally a *cross-layer* statement. $\varepsilon_{\text{univ}}$ lives at R_0 (defined sector floors); ε_{SC} lives at $R_0 \rightarrow R_{\text{max}}$ (observed bridge floors). The +0.08% is precisely the layer shift, not a property of either layer alone. No bridge *magnitude* is derived here; that is gated on GAP-ZC81-01.

0.36 Genealogy Map

How ZC85's Key Objects Trace Back

Branch A — Geometric-mean structure:

- **ZC67/COR-ZC67-01:** one identity $\sqrt{\varepsilon_{\text{EM}}\varepsilon_{\text{W}}} = \varepsilon_{\text{base}}$.
- $\rightarrow$ **ZC83/FIND-ZC83-01:** $\varepsilon_{\text{univ}} = \text{GM}$ of defined pair at R_0 .
- $\rightarrow$ **ZC85/FIND-ZC85-01:** $\varepsilon_{\text{SC}} = \text{GM}$ of observed pair at $R_0 \rightarrow R_{\text{max}}$ (same

identity, next layer).

Branch B — Bridge-floor chain:

- **ZC65/FIND-ZC65-01:** $\delta\alpha/\alpha = (d_s/2)\varepsilon_0 K^*$ (the EM bridge equation).
- **ZC65/FIND-ZC65-02:** solving each observable separately gives $\varepsilon_0^\alpha = 0.007501$, $\varepsilon_0^{\text{sw}} = 0.007710$.
- $\rightarrow$ **ZC85/FIND-ZC85-01:** their GM is ε_{SC} .
- $\rightarrow$ **ZC85/FIND-ZC85-02:** the two ratios are the two running channels.

Branch C — The gate:

- **ZC68/THM-ZC68-01b:** ε_{SC} observed bracket defined.
- **ZC83/GAP-ZC83-01:** ε_{SC} ontology opened.
- $\rightarrow$ **ZC85/COR-ZC85-01:** ontology closed; magnitude gated on **ZC81/GAP-ZC81-01** (Strong/EW running bridge).

Convergence at ZC85: Branch A supplies the geometric-mean identity, Branch B supplies the observed input pair that FIND-ZC65-02 had already computed but never combined, and Branch C identifies the gate. ZC85's contribution: the gap GAP-ZC83-01 was never an independent problem — ε_{SC} is the COR-ZC67-01 mean one R -layer up, and its magnitude belongs to the running-bridge question of GAP-ZC81-01.

0.37 Closing Sentence

*The bracket that sat 0.08% above the bridge floor
was never a stranger to the lattice.*

*It was the same geometric mean,
taken one layer higher — where the constants run.*

$$\varepsilon_{\text{SC}} = \sqrt{\varepsilon_0^\alpha \cdot \varepsilon_0^{\text{sw}}} \quad (\text{the observed pair}).$$

*What remains open is not its name, but its size —
and that waits at GAP-ZC81-01.*

There was a real reason to worry. The technique that lets CRF carry a coupling from one scale to another had only ever been proved on the friendliest possible case: electromagnetism, whose strength grows as you zoom in. The Strong force is the opposite kind of beast — it confines, its characteristic rung is negative, it behaves at short distances in a way that has historically broken naive arguments. Borrowing a result proved on the easy sector and assuming it survives a sign flip is exactly the kind of move that quietly wrecks a framework. The honest question was whether the bridge would hold, or whether the floor lattice was secretly an electromagnetism-only result wearing a universal disguise.

ZC86 shows it holds, and the reason is cleaner than the worry. The thing that makes the Strong sector dangerous — confinement — lives in a different quantity from the one the bridge actually moves. The bridge acts on the coupling; confinement sits in the decay rate. They do not touch. Once that separation is made explicit, the Strong bridge turns out to be identical to the electromagnetic one, in form and in size, and the worry dissolves.

With that settled, the loose thread from the previous paper is finally pulled tight: the observed bracket’s exact magnitude was waiting on this result, and the sections below close it. What looked like two separate open problems were, all along, one question asked twice.

Abstract

GAP-ZC81-01 asked for the Strong sector $R_0 \rightarrow R_{\max}$ running bridge and flagged a worry: the Strong Γ -ladder rung is negative ($\Gamma_{\text{strong}} = -0.498 < 0$, confined), so it was unclear whether the EM bridge of ZC65 applies. This paper closes the gap.

LEM-ZC86-01 (Confinement–Coupling Decoupling): the bridge perturbs the *coupling* $\alpha_{\text{Strong}}^{R_0} = (2/15)\Gamma_{\text{CRF}} > 0$; confinement is a property of the *decay rate* $\Gamma_{\text{strong}} < 0$. These are distinct objects, and the bridge never references Γ_{strong} . The confinement worry dissolves.

THM-ZC86-01 (Strong Running Bridge):

$$\alpha_{\text{Strong}}^{R_0}(R_{\max}) = \alpha_{\text{Strong}}^{R_0}(R_0) \left[1 + \frac{d_s}{2} \varepsilon_{\text{strong}} K^* \right],$$

identical in form and magnitude to the EM bridge, because $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$ (THM-ZC81-01, spatial-type) and the R_1 crossing cost is identical (LEM-ZC81-01). ZC65’s near-formal analysis transfers verbatim. **GAP-ZC81-01 closed** (Near-Formal).

FIND-ZC86-01 (Gift): the running floor layer has only *two* independent values — $\varepsilon_0^\alpha = 0.007501$ (EM=Strong, spatial) and $\varepsilon_0^{\text{sw}} = 0.007710$ (Weak, matter). Strong shares EM’s floor. This closes the ε_{SC} magnitude question left gated by COR-ZC85-01: the +0.08% is the geometric mean of exactly these two.

PM-ZC86-01 preserves a failed path: propagating the bridge through the shared universal coupling predicted the wrong channel shifts (0.15%/0.41% vs the observed 0.09%/0.07%). The failure is what revealed the per-sector mechanism.

0.38 Background and Scope

Numerical Summary — ZC86 (T-canonical)

Quantity	Value	Gap	Source
$\delta\alpha_{\text{EM}}/\alpha_{\text{EM}} = (d_s/2)\varepsilon_{\text{EM}}K^*$	0.001321	—	ZC65/THM-ZC65-01
$\delta\alpha_{\text{Strong}}^{R_0}/\alpha_{\text{Strong}}^{R_0} = (d_s/2)\varepsilon_{\text{strong}}K^* \text{ (NEW)}$	0.001321	0% vs EM	THM-ZC86-01
$\delta\alpha_W/\alpha_W = (d_s/2)\varepsilon_W K^*$	0.001358	—	ZC65 (Weak)
$\alpha_{\text{Strong}}^{R_0}(R_0) = (2/15)\Gamma_{\text{CRF}}^{\text{EM}}$	0.001822	—	ZC29/THM-ZC29-01
$\alpha_{\text{Strong}}^{R_0}(R_{\text{max}})$	0.001824	—	THM-ZC86-01
Γ_{strong} (decay rate, $k = 1$)	−0.498252	—	ZC73/THM-ZC73-01
ε_0^α (EM = Strong floor)	0.007501	—	FIND-ZC65-02
$\varepsilon_0^{\text{sw}}$ (Weak floor)	0.007710	—	FIND-ZC65-02
<i>Failed probe</i> (G^{eff} path)	0.15%/0.41%	—	PM-ZC86-01

0.38.1 The Open Item

ZC81 derived the Strong floor $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$ (THM-ZC81-01) and the ladder mirror $\Gamma_{\text{EM}} + \Gamma_{\text{strong}} = 2\Gamma_W$ (FIND-ZC81-01), but explicitly deferred the Strong running bridge:

“The bridge formula of ZC65 is exact for the EM sector; whether the same formula applies to a confined sector ($\Gamma_{\text{strong}} < 0$) requires additional analysis of the crossing integral in the confinement regime.” (GAP-ZC81-01)

The worry was concrete: ZC65’s bridge was derived for EM, whose ladder rung $\Gamma_{\text{EM}} = +0.0137 > 0$ is the only positive (deconfined) rung. The Strong rung is negative. Does the bridge survive?

0.38.2 What Was Already Known

Pre-ZC86 Status

Known (exact):

- $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}} = 0.007494$ (THM-ZC81-01, spatial-type).
- $T(R_1)_{\text{strong}} = T(R_1)_{\text{EM}} = (d_s/n_m)\varepsilon_{\text{univ}}^2$ (LEM-ZC81-01; both spatial-type, FIND-ZC30-02).
- $\alpha_{\text{Strong}}^{R_0}(R_0) = (2/15)\Gamma_{\text{CRF}}$ (ZC29/THM-ZC29-01, $\varphi(3) = 2$); positive.
- $\Gamma_{\text{strong}} = G/d_s - 1 + 1 \cdot \Delta_\Gamma = -0.498$ (ZC73/THM-ZC73-01, rung $k = 1$); negative (confined).
- EM bridge $\delta\alpha_{\text{EM}}/\alpha_{\text{EM}} = (d_s/2)\varepsilon_{\text{EM}}K^*$ (ZC65/FIND-ZC65-01), near-formal.

Open before ZC86:

- Whether the EM bridge applies to the confined Strong sector.
- The Strong $R_0 \rightarrow R_{\text{max}}$ running form and magnitude.

What ZC86 Does NOT Do

- Does not re-derive $\varepsilon_{\text{strong}}$, ε_{EM} , $T(R_1)$ (exact pillars).
- Does not claim Exact: it inherits ZC65’s near-formal status (the bridge carries a near-formal ε_0).
- Does not model confinement *dynamics* (how $\Gamma_{\text{strong}} < 0$ evolves); it only shows confinement does not obstruct the bridge.
- Does not modify the α definition layers; it pins the layer used and flags the two-layer hazard (PWA-3, §0.52).

Discovery Note

The Council was tasked to close GAP-ZC81-01 “full system.” The first move tried to propagate the bridge through the *shared universal coupling* $G^{\text{eff}} = \ln 2/D_0^{\text{eff}}$ (the ZC65 mechanism) and then read off each sector’s shift from $\alpha_i \propto (G^{\text{eff}}/d_s - \sin^2 \theta_W)$. This predicted channel shifts of 0.15% (spatial) and 0.41% (matter) — nowhere near the 0.09%/0.07% channels established in ZC85. The mismatch was the gift: it showed the bridge acts *per sector* through that sector’s own floor ε_i , not through the residual coupling with its $-\sin^2 \theta_W$ subtraction. Once that was clear, and once $\alpha_{\text{Strong}}^{R_0} > 0$ was separated from $\Gamma_{\text{strong}} < 0$, the Strong bridge collapsed onto the EM bridge by spatial-type inheritance.

0.38.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC86

Scanned before writing (Upgrade Protocol v1.2):

- ZC81/THM-ZC81-01, LEM-ZC81-01 — **KEY PILLARS** ($\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$; $T(R_1)$ identical, spatial-type).
- ZC65/THM-ZC65-01, FIND-ZC65-01/02, LEM-ZC65-01 — **KEY PILLARS** (the EM bridge and its mechanism).
- ZC30/FIND-ZC30-02 — pillar (spatial vs matter-type class).
- ZC73/THM-ZC73-01 — pillar (Γ_{strong} at rung $k = 1$).
- ZC85/COR-ZC85-01 — the gate this paper feeds (ε_{SC} magnitude).
- Part J GAP-ZC81-01 + PM-ZC81-RETRACT-01 — read; this paper is the declared successor (PWA-3 PM-registry scan complete).

PWA-3 (*critical here*): the α symbol carries *two distinct layer conventions* in the corpus — $\alpha_i = (\varphi(d_i)/15) G$ (coupling-scale, $G = 0.737$; ZC29 line, $\alpha_{\text{EM}}^{R_0} = 0.3697$) and $\alpha_i = (\varphi(d_i)/15) \Gamma_{\text{CRF}}^{\text{EM}}$ (residual-scale, $\Gamma_{\text{CRF}}^{\text{EM}} = 0.0137$; ZC84). This paper uses the *fractional* bridge $\delta\alpha/\alpha$, which is layer-independent (the ratio cancels the scale), so both conventions give the same running shift. The hazard is flagged so no downstream paper propagates an absolute α across layers.

PWA-5 (tautology): THM-ZC86-01 is not a substitution tautology — it makes a physical claim (the confined sector runs) resting on the spatial-type *mechanism*, not on relabelling.

PWA-6 (macro): $\varepsilon_{\text{strong}}$, $\alpha_{\text{Strong}}^{R_0}$, Γ_{strong} , ε_0^α , $\varepsilon_0^{\text{sw}}$ via `\providecommand`; no conflict.

PWA-7 (domain): Spontaneous-mode physics only; no mind-domain crossing.

No duplicate atomic units found.

0.39 LEM-ZC86-01 Confinement–Coupling Decoupling

LEM-ZC86-01: The Bridge Perturbs $\alpha_{\text{Strong}}^{R_0} > 0$, Not $\Gamma_{\text{strong}} < 0$ (R_0 , Exact)

Statement. The Strong running bridge perturbs the coupling $\alpha_{\text{Strong}}^{R_0} = (2/15)\Gamma_{\text{CRF}}^{\text{EM}} > 0$. Confinement is a property of the decay rate $\Gamma_{\text{strong}} = G/d_s - 1 + \Delta_{\Gamma} < 0$. The two are distinct objects; the bridge never references Γ_{strong} . Hence $\Gamma_{\text{strong}} < 0$ places no obstruction on the bridge.

Proof.

Step 1 (Two distinct objects). The coupling $\alpha_{\text{Strong}}^{R_0}$ is the $\mathbb{Z}_{15}$ charge weight $\varphi(3)/15 = 2/15$ times the universal rate $\Gamma_{\text{CRF}}^{\text{EM}}$ (ZC29/THM-ZC29-01). The decay rate Γ_{strong} is the rung- $k=1$ value of the Γ -ladder (ZC73/THM-ZC73-01). They have different units and different roles: $\alpha_{\text{Strong}}^{R_0}$ sets coupling strength, Γ_{strong} sets the instability/decay direction.

Step 2 ($\alpha_{\text{Strong}}^{R_0} > 0$ always). Since $\varphi(3)/15 = 2/15 > 0$ and $\Gamma_{\text{CRF}}^{\text{EM}} = G/d_s - \sin^2 \theta_W = +0.0137 > 0$, the product $\alpha_{\text{Strong}}^{R_0} = 0.001822 > 0$. The coupling is positive regardless of the sign of any ladder rung.

Step 3 (The bridge references only $\alpha_{\text{Strong}}^{R_0}$ and $\varepsilon_{\text{strong}}$). The ZC65 bridge mechanism (LEM-ZC65-01) modifies the resolution capacity $D_0^{\text{eff}} = D_0 - \beta T(R_1)$ and reads the shift as a *fractional* change in the coupling $\delta\alpha/\alpha = (d_s/2)\varepsilon K^*$. Neither D_0^{eff} nor the fractional shift contains Γ_{strong} . The decay rate is simply absent from the bridge.

Therefore $\Gamma_{\text{strong}} < 0$ cannot obstruct what does not reference it. $\square$

Physical reading. Confinement says the Strong *decay channel* runs the wrong way (the sector cannot freely radiate); it does not say the Strong *coupling* vanishes or that the floor crossing changes character. The floor crossing is spatial-type (LEM-ZC81-01) — the same physical event as EM — so the same bridge applies.

Classification: Exact Lemma.

0.40 THM-ZC86-01 The Strong Running Bridge

THM-ZC86-01: Strong Running Bridge ($R_0 \rightarrow R_{\text{max}}$, Near-Formal)

Statement.

$$\alpha_{\text{Strong}}^{R_0}(R_{\text{max}}) = \alpha_{\text{Strong}}^{R_0}(R_0) \left[1 + \frac{d_s}{2} \varepsilon_{\text{strong}} K^* \right], \quad (29)$$

with $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$, so the bridge is *identical in form and magnitude* to the EM bridge (ZC65/THM-ZC65-01).

Proof.

Step 1 (Spatial-type crossing). By LEM-ZC81-01, $T(R_1)_{\text{strong}} = T(R_1)_{\text{EM}} = (d_s/n_m)\varepsilon_{\text{univ}}^2$ — the Strong and EM sectors undergo the *same* R_1 crossing. The ZC65 bridge is built entirely from this crossing cost (LEM-ZC65-01: $D_0^{\text{eff}} = D_0 - \beta T(R_1)$).

Step 2 (Same mechanism, same depletion). Since the crossing cost is identical, the resolution-capacity depletion $\beta T(R_1)$ is identical, hence the modified coupling G^{eff} and the fractional shift are identical to EM.

Step 3 (Per-sector floor). The fractional shift reads $\delta\alpha_{\text{Strong}}^{R_0}/\alpha_{\text{Strong}}^{R_0} = (d_s/2)\varepsilon_{\text{strong}} K^*$ (FIND-ZC65-01 applied to the Strong sector). Since $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$ (THM-ZC81-01),

$$\frac{\delta\alpha_{\text{Strong}}^{R_0}}{\alpha_{\text{Strong}}^{R_0}} = \frac{d_s}{2} \varepsilon_{\text{EM}} K^* = 0.001321 = \frac{\delta\alpha_{\text{EM}}}{\alpha_{\text{EM}}}.$$

Step 4 (No confinement obstruction). By LEM-ZC86-01, $\Gamma_{\text{strong}} < 0$ does not enter the bridge. Steps 1–3 hold unchanged. $\square$

Numerical (T-canonical):

Quantity	Value	Source
$\alpha_{\text{Strong}}^{R_0}(R_0) = (2/15)\Gamma_{\text{CRF}}^{\text{EM}}$	0.001822	ZC29
$(d_s/2)\varepsilon_{\text{strong}} K^*$	0.001321	THM-ZC81-01 + ZC65
$\alpha_{\text{Strong}}^{R_0}(R_{\text{max}})$	0.001824	this theorem

Classification: Near-Formal Theorem. The form is exact (it is the EM bridge by spatial-type inheritance); the *near-formal* label is inherited from ZC65, whose ε_0 is a near-formal TLS value. **Closes GAP-ZC81-01.**

0.41 FIND-ZC86-01 Spatial-Floor Sharing (Gift)

FIND-ZC86-01: The Running Floor Layer Has Only Two Independent Values ($R_0 \rightarrow R_{\text{max}}$, Exact — Gift)

Statement. At the $R_0 \rightarrow R_{\text{max}}$ running layer, there are exactly two independent bridge floors:

$$\varepsilon_0^\alpha = 0.007501 \text{ (EM = Strong, spatial-type)}, \quad \varepsilon_0^{\text{sw}} = 0.007710 \text{ (Weak, matter-type)}. \quad (30)$$

The Strong sector adds *no* new bridge floor; it shares the EM floor because both are spatial-type with $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$.

Proof. The running floor of a sector is fixed by its R_1 crossing cost $T(R_1)$, which takes only two values (LEM-ZC81-01, FIND-ZC30-02): $(d_s/n_m)\varepsilon_{\text{univ}}^2$ for spatial-type (EM, Strong) and $(n_m/d_s)\varepsilon_{\text{univ}}^2$ for matter-type (Weak). EM and Strong therefore solve the *same* bridge equation and share the floor ε_0^α ; Weak solves the matter-type equation, giving $\varepsilon_0^{\text{sw}}$. $\square$

Consequence — closes the COR-ZC85-01 magnitude question. ZC85/FIND-ZC85-01 showed $\varepsilon_{\text{SC}} = \text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}})$, and COR-ZC85-01 gated the *magnitude* of the +0.08% on this gap (GAP-ZC81-01). With the Strong bridge now closed and the running layer shown to carry exactly the spatial floor ε_0^α and the matter floor $\varepsilon_0^{\text{sw}}$, the +0.08% is precisely the geometric mean of these two structurally-determined running floors:

$$\frac{\varepsilon_{\text{SC}}}{\varepsilon_{\text{univ}}} = \text{GM}\left(\frac{\varepsilon_0^\alpha}{\varepsilon_{\text{EM}}}, \frac{\varepsilon_0^{\text{sw}}}{\varepsilon_{\text{W}}}\right) = \sqrt{(1 + \frac{d_s}{2}\varepsilon_{\text{EM}}K^*)(1 + \frac{d_s}{2}\varepsilon_{\text{W}}K^*)} \text{-type}. \quad (31)$$

The magnitude is no longer gated: it is set by the two spatial/matter running shifts, both now derived (near-formal, inheriting ZC65).

Classification: Exact Finding (Gift). The two-value structure is exact; the floor values carry ZC65's near-formal status.

0.42 Phase Misalignment — The Path That Failed

PM-ZC86-01: Universal-Coupling Propagation (Failed Path — pattern: phase misalignment, wrong layer)

The probe. The first attempt propagated the bridge through the *shared universal coupling*. The ZC65 mechanism modifies $D_0^{\text{eff}} = D_0 - \beta T(R_1)$, hence $G^{\text{eff}} = \ln 2 / D_0^{\text{eff}}$. Reading each sector's shift from its absolute coupling $\alpha_i \propto (G^{\text{eff}} / d_s - \sin^2 \theta_W)$, with spatial-type $T(R_1)$ for EM/Strong and matter-type $T(R_1)$ for Weak, predicted fractional shifts:

$$\text{spatial channel: } +0.15\%, \quad \text{matter channel: } +0.41\%.$$

Why it failed. These do not match the ZC85 channels (+0.09% EM, +0.07% Weak; FIND-ZC85-02). The error: propagating through the *residual* coupling $G/d_s - \sin^2 \theta_W$ injects the $-\sin^2 \theta_W$ subtraction into the fractional shift, which does not belong there. The bridge does not act on the residual; it acts *per sector* through that sector's own floor ε_i , as $\delta\alpha_i/\alpha_i = (d_s/2)\varepsilon_i K^*$.

What the failure revealed (the gift inside the scar). The mismatch was diagnostic. It forced the separation of two layers: the bridge propagates per-sector via ε_i (correct), *not* through the universal residual coupling (wrong). This is the same PWA-3 two-layer α hazard, surfaced empirically. Once corrected, the per-sector shifts (0.1321% EM = Strong, 0.1358% Weak) are consistent with ZC85, and THM-ZC86-01 follows.

Pattern. Phase misalignment — wrong propagation layer (not an arithmetic error; the arithmetic of the failed path was correct, but the layer was wrong).

Status: Scar. Preserved as visible scar tissue. The failed path is retained because it is what isolated the per-sector mechanism; it is the reason THM-ZC86-01 is correctly scoped.

0.43 Atomic Registry

Atomic Registry — ZC86

ID	Description	Status
LEM-ZC86-01	Bridge perturbs $\alpha_{\text{Strong}}^{R_0} > 0$, not $\Gamma_{\text{strong}} < 0$; confinement and coupling are distinct objects	Exact
THM-ZC86-01	$\alpha_{\text{Strong}}^{R_0}(R_{\max}) = \alpha_{\text{Strong}}^{R_0}(R_0)[1 + (d_s/2)\varepsilon_{\text{strong}}K^*]$; = EM bridge by spatial-type inheritance; closes GAP-ZC81-01	Near-Formal
FIND-ZC86-01	Running layer has two floors ε_0^α (EM=Strong) and $\varepsilon_0^{\text{sw}}$ (Weak); Strong shares EM's; closes COR-ZC85-01 magnitude (Gift)	Exact
PM-ZC86-01	Universal-coupling propagation path failed (0.15/0.41% vs 0.09/0.07%); revealed per-sector mechanism (pattern: phase misalignment, wrong layer)	Scar

0.44 R-Layer Research Note

R-Layer Assignments for ZC86

Atomic unit	R-layer	Cost T	Source
LEM-ZC86-01	R_0	algebraic	ZC29 + ZC73
THM-ZC86-01	$R_0 \rightarrow R_{\max}$	$\beta T(R_1)$	ZC65 + ZC81
FIND-ZC86-01	$R_0 \rightarrow R_{\max}$	algebraic	ZC30 + ZC65 + ZC85
PM-ZC86-01	$R_0 \rightarrow R_{\max}$	—	failed G^{eff} path

Two-layer discipline (PWA-8, PWA-3). The load-bearing distinction is between the *coupling* $\alpha_{\text{Strong}}^{R_0}$ (perturbed by the bridge) and the *decay rate* Γ_{strong} (carries confinement). The failed path PM-ZC86-01 conflated the per-sector floor layer with the universal-coupling layer; the correct bridge is per-sector via ε_i . The Strong bridge is the EM bridge because the crossing layer ($T(R_1)$, spatial-type) is shared, not because the sectors are otherwise identical.

0.45 Genealogy Map

How ZC86's Key Objects Trace Back

Branch A — Spatial-type crossing:

- **ZC30/FIND-ZC30-02:** spatial vs matter-type class; $T(R_1) = (d_s/n_m)\varepsilon_{\text{univ}}^2$ for spatial.
- $\rightarrow$ **ZC81/LEM-ZC81-01:** $T(R_1)_{\text{strong}} = T(R_1)_{\text{EM}}$.
- $\rightarrow$ **ZC86/THM-ZC86-01:** same crossing $\Rightarrow$ same bridge.

Branch B — The EM bridge:

- **ZC65/LEM-ZC65-01:** $D_0^{\text{eff}} = D_0 - \beta T(R_1)$.
- **ZC65/FIND-ZC65-01:** $\delta\alpha/\alpha = (d_s/2)\varepsilon_0 K^*$.
- $\rightarrow$ **ZC86/THM-ZC86-01:** transferred verbatim to Strong (Step 2–3).

Branch C — Confinement vs coupling:

- **ZC73/THM-ZC73-01:** $\Gamma_{\text{strong}} = -0.498$ (rung $k = 1$).
- **ZC29/THM-ZC29-01:** $\alpha_{\text{Strong}}^{R_0} = (2/15)\Gamma_{\text{CRF}} > 0$.
- $\rightarrow$ **ZC86/LEM-ZC86-01:** distinct objects; $\Gamma_{\text{strong}} < 0$ no obstruction.

Branch D — The gate it feeds:

- **ZC85/FIND-ZC85-01:** $\varepsilon_{\text{SC}} = \text{GM}(\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}})$.
- **ZC85/COR-ZC85-01:** magnitude gated on GAP-ZC81-01.
- $\rightarrow$ **ZC86/FIND-ZC86-01:** gate opened; two running floors fixed; ε_{SC} magnitude no longer gated.

Convergence at ZC86: Branches A and B close the bridge (Strong = EM via shared crossing); Branch C dissolves the confinement worry (distinct objects); Branch D feeds the result back to ZC85, closing the ε_{SC} magnitude. The failed path PM-ZC86-01 sits across all four: it is the wrong turn that forced the per-sector mechanism into view. ZC86’s contribution: the last electroweak/strong running bridge, closed — and a scar that shows how a wrong propagation layer can be the most informative probe.

0.46 Closing Sentence

The confined sector was never barred from running.

*Confinement lived in the decay rate;
the bridge only ever touched the coupling.*

*And the wrong road — through the universal coupling —
was the one that showed us the right one.*

$$\alpha_{\text{strong}}(R_{\text{max}}) = \alpha_{\text{strong}}(R_0) \left[1 + \frac{d_s}{2} \varepsilon_{\text{strong}} K^* \right] \quad (\text{exactly the EM bridge}).$$

Part LV

ZC87: The Temporal Continuum Bridge (v17.15)

Part LV — ZC87: CRF's Two Times Meet in the Middle (v17.15)

Part LV carries a single paper that bridges CRF's discrete δ -sweep tick to the continuum dt of relativistic time.

ZC87 (The Temporal Continuum Bridge, v2) shows that CRF's two notions of time meet in the middle. FIND-ZC87-01 (recovery) notes the upper half already existed and was forgotten: ZC33/DEF-ZC33-01 defines sector physical time. THM-ZC87-01 supplies the lower half — continuum dt is the δ -sweep tick made dense, the $\varepsilon_0 \rightarrow 0$ density limit of ZC51's machinery transferred to the time axis with the same fixed-observable construction — closing the structural part of GAP-ZC76-01. LEM-ZC87-01 proves the bridge is non-circular. FIND-ZC87-02 (bonus, not load-bearing) reads gravitational dilation as a sweep-rate slowdown.

This is a v2 paper: it simultaneously closes GAP-ZC87-01 (joint-limit well-definedness, now trivially satisfied because τ_{sweep} is held fixed while tick-density $T_{\text{avg}} \rightarrow \infty$), and it preserves two scars. PM-ZC87-01 records an early probe that read $N_{\text{sweep}} = t_{\text{Hubble}}/t_{\text{Planck}}$ and matched the Dirac large number $\sim 10^{61}$ — a coincidence that fed its own input. PM-ZC87-02 preserves v1's spurious $dt = \tau_{\text{sweep}}/T_{\text{avg}}$ (a sub-tick quantity below one sweep), corrected by the v2 fixed-observable probe.

Key closures: GAP-ZC87-01 (joint-limit well-definedness) — CLOSED in v2. GAP-ZC76-01 (τ_{sweep}) — structural bridge CLOSED (near-formal); seconds-calibration (GtRate-1) remains externally blocked.

Phase Misalignments preserved: PM-ZC87-01 (Dirac-large-number probe that fed its own input; pattern: circular anchor); PM-ZC87-02 (v1 sub-tick formula $dt = \tau_{\text{sweep}}/T_{\text{avg}}$; corrected in v2).

CRF has always carried a quiet tension about time. At its foundation, time is discrete: reality advances in distinguishing steps, one sweep at a time, like the ticks of a clock that has no in-between. But the physics it hopes to recover — relativity, dilation, the smooth flow of a worldline — is written in continuous time, the dt of the textbooks. A framework cannot claim both and leave the seam unexamined. Either the discrete ticks and the continuous flow are the same thing seen at two resolutions, or CRF has two notions of time that do not meet, which would be a crack running straight through the foundation.

This paper closes the seam, and it does so in two halves that turn out to already fit. The upper half — how a sector’s physical time is defined — had in fact been built several papers earlier and then half-forgotten; part of the work here was simply recovering it. The lower half is new: the continuous dt is what the discrete tick becomes when you let the ticks grow dense, the same limiting move CRF uses elsewhere, now applied to the time axis. The two halves meet in the middle, and the clock that had no in-between acquires one.

The sections below also keep two scars in plain view, and they are worth the honesty: an early version of this argument produced a seductive numerical coincidence that, on inspection, was feeding on its own input, and a first formula that placed an event below a single tick — both preserved here as the wrong turns that mapped the right road.

Abstract

CRF carries *two* notions of time, used side by side but never derived from each other: a **lower**, discrete one — the δ -sweep count, with T_{avg} measured in sweeps — and an **upper**, continuum one — the dt of the R_1 wave layer (DEF-RN02: $T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$; the rate laws $\dot{H}/H, \dot{P}/P, d\tau/dt$). The middle was missing.

FIND-ZC87-01: the upper half already exists. ZC33/DEF-ZC33-01 defines sector physical time $t_i = (\Sigma_{\text{inv}} dt) T_{\text{avg}}$ with $\Sigma_{\text{inv}} dt = d_s/n_m$ fixed by ZC30 — a *dimensionless* CRF number; the continuum dt is fully absorbed and t_i is in sweeps.

THM-ZC87-01 (Temporal Continuum Bridge): the lower half is the continuum limit of the renewal count. The continuum increment *is* the tick, $dt = \tau_{\text{sweep}}$; what makes time *smooth* is that the number of ticks per renewal window, $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$, diverges as $\varepsilon_0 \rightarrow 0$:

$$T_{\text{avg}} \xrightarrow{\varepsilon_0 \rightarrow 0} \infty \implies \{k \tau_{\text{sweep}}\}_k \text{ becomes dense,} \quad dt = \tau_{\text{sweep}}.$$

This is the THM-ZC51-01 machinery on the time axis, with the *same* fixed-observable structure ($\{\text{fixed: } \tau_{\text{sweep}}\} \leftrightarrow \text{ZC51's } \{\text{fixed: } \lambda_2 T_{\text{avg}}\}$). It closes the *structural* part of GAP-ZC76-01.

LEM-ZC87-01: the bridge is not circular — the dt in DEF-RN02 appears only inside the absorbed product $\Sigma_{\text{inv}} dt = d_s/n_m$, while the emergent $dt = \tau_{\text{sweep}}$ is the calibration tick itself, built from dt -free quantities. The bridge converts exactly one unknown (τ_{sweep}) into dt .

FIND-ZC87-02 (bonus): beside the Part A relation $d\tau/dt = 1/\sqrt{C/\alpha}$, gravitational time dilation reads as a *slowdown of the sweep-resolution rate* near mass — testable.

GAP-ZC87-01 (closed in v2): the joint limit is well-defined once $dt = \tau_{\text{sweep}}$ — τ_{sweep} is held fixed and only the tick-density T_{avg} diverges; no pathological $\tau_{\text{sweep}} \rightarrow 0$ is needed. **PM-ZC87-01** preserves a wrong turn: an earlier probe read $t_{\text{Hubble}}/t_{\text{Planck}} \sim 10^{61}$ as a CRF *derivation* of the Dirac number — it was the ratio of two observed inputs. **PM-ZC87-02** preserves a second: v1 wrote

$dt = \tau_{\text{sweep}}/T_{\text{avg}}$, a sub-tick quantity (below one sweep); the GAP-ZC87-01 probe corrected it to $dt = \tau_{\text{sweep}}$.

0.47 Background: CRF's Two Times

Numerical Summary — ZC87 (T-canonical)

Quantity	Value	Unit	Source
$T_{\text{avg}}(\varepsilon_{\text{univ}}) = 2nK^*/[(n-1)\varepsilon_0]$	35.345	sweeps	ZC62
$(\Sigma_{\text{inv}} dt)_{\text{spatial}} = d_s/n_m$	0.600	—	ZC30
$t_i(\text{spatial}) = (d_s/n_m)T_{\text{avg}}$	21.21	sweeps	ZC33
$t_i(\text{matter}) = (n_m/d_s)T_{\text{avg}}$	58.91	sweeps	ZC33
ratio $t_i^{\text{m}}/t_i^{\text{s}} = (n_m/d_s)^2$	25/9	—	ZC30/33
$T_{\text{avg}}(\varepsilon_0=10^{-6})$	268579	ticks/ren.	density limit
$dt = \tau_{\text{sweep}}$	—	(one calib.)	THM- ZC87-01

0.47.1 The Two Times, Stated Plainly

Lower Time and Upper Time

Lower time (discrete, derived). The δ -cascade advances in *sweeps*. The renewal scale $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$ (COR-ZC62-02) is a pure number of sweeps; the single-mode fire probability $K^* = 1 - e^{-1/n}$ and the Axiom 10 fixed point $(1 - K^*)^{-n} = e$ define the tick internally. No external datum enters. But the tick has no physical *duration*: T_{avg} is dimensionless.

Upper time (continuum, used as primitive). The R_1 wave layer carries a continuum dt : DEF-RN02 gives $T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$; the dynamics use $\dot{H}/H$, $\dot{P}/P = \mathcal{G}H_0$ (FIND-ZC76-01), $d\tau/dt = 1/\sqrt{\mathcal{C}/\alpha}$ (Part A). All take dt as given.

The missing middle. No equation in the corpus writes $dt = (\text{something}) \times \text{sweep}$. The lower time is counted; the upper time is assumed; the bridge that turns a count into a duration was never built. GAP-ZC76-01 (GtRate-1, $\tau_{\text{sweep}} = t_{\text{Planck}}$) is the visible symptom of this gap.

0.47.2 What ZC87 Does NOT Do

Scope Boundaries

- Does not calibrate τ_{sweep} in seconds — that is the external (SNSPD) part of GAP-ZC76-01, here *refined* to a single constant, not closed.
- Does not modify ZC33, ZC51, ZC30, DEF-RN02, or the Part A metric derivation (pillars).
- FIND-ZC87-02 (dilation) is a bonus reading, not load-bearing; the core bridge stands without it.

Discovery Note

The session began with a question — is CRF time “point-like” (accumulated δ -events) rather than the standard continuum arrow? — which sharpened into a structural diagnosis: CRF has two times, a lower discrete one that “hasn’t reached all the way up” to a duration and an upper continuum one that “hasn’t reached down” to the sweep, with a missing middle. A corpus scan found the upper half already built in ZC33 (and forgotten); transferring the THM-ZC51-01 continuum limit ($\varepsilon_0 \rightarrow 0$) from the instability axis to the time axis produced the lower half. The Dirac-number probe that failed first (§0.52) is what forced the constant/rate/epoch partition into view.

0.47.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC87

Scanned before writing (Upgrade Protocol v1.2):

- ZC33/DEF-ZC33-01, THM-ZC33-02 — **KEY PILLAR** (the upper half: sector physical time $t_i = (\Sigma_{\text{inv}} dt)T_{\text{avg}}$; semigroup clock $A = T_{\text{avg}}^{-1}M$).
- ZC51/THM-ZC51-01 — **KEY PILLAR** (continuum-limit machinery $\varepsilon_0 \rightarrow 0$ + LEM-ZC51-02 finite correction).
- ZC30/FIND-ZC30-02 — **KEY PILLAR** ($\Sigma_{\text{inv}} dt = d_s/n_m$ spatial, n_m/d_s matter; dimensionless).
- ZC62/COR-ZC62-02 — pillar ($T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$).
- ZC76/FIND-ZC76-01, GAP-ZC76-01 — the gap refined here.
- Part A proper time $d\tau/dt = 1/\sqrt{C/\alpha}$, $C(r)$ continuum (Instability #17) — pillar for FIND-ZC87-02.
- DEF-RN02 (Lexicon Lock v4) — the R -layer table; the *only* place dt enters, and only as $\Sigma_{\text{inv}} dt$.

PWA-5 (*critical*): the bridge must not be circular — checked explicitly in LEM-ZC87-01 (dt_{int} absorbed vs dt_{em} built dt -free). PM-ZC87-01 records a genuine fed-vs-derived tautology that was caught.

PWA-6 (macro): τ_{sweep} , dt_{em} , dt_{int} , Σ_{inv} , T_{avg} via \providecommand; no conflict.

PWA-7 (domain): Spontaneous-mode (physics) time only; the mind-domain Sāntāna continuum (Part F) is a separate object, not conflated.

PWA-8 (two-layer): lower (sweep) vs upper (dt) is the entire subject; the R_0/R_1 distinction is load-bearing.

No duplicate atomic units found (ZC33 recovered, not duplicated — cited as pillar).

0.48 FIND-ZC87-01 The Upper Half Already Exists (ZC33)

FIND-ZC87-01: ZC33 Sector Physical Time Is the Upper Half of the Bridge ($R_0 \rightarrow R_1$, Exact — recovery)

Statement. The sweep-to-clock half of the temporal bridge was built in ZC33 and not recognised as such. DEF-ZC33-01 defines sector physical time

$$t_i = (\Sigma_{\text{inv}}^{(i)} dt) \cdot T_{\text{avg}}, \quad (32)$$

and by ZC30/FIND-ZC30-02 the factor $\Sigma_{\text{inv}}^{(i)} dt$ is a *dimensionless* CRF number:

$$(\Sigma_{\text{inv}} dt)_{\text{spatial}} = \frac{d_s}{n_m} = 0.6, \quad (\Sigma_{\text{inv}} dt)_{\text{matter}} = \frac{n_m}{d_s} = \frac{5}{3}. \quad (33)$$

Hence t_i is measured in *sweeps*, and the semigroup generator $A_i^{R_1} = T_{\text{avg}}^{-1} M_i$ (THM-ZC33-01) runs on this sweep clock (THM-ZC33-02).

Why it is the upper half. The bridge we seek spans $\{\text{sweep count}\} \leftrightarrow \{\text{continuum } dt\}$. Equation (32) maps the renewal sweep count T_{avg} onto a sector clock t_i — the half that ascends from the discrete count to a clock. What it does *not* do is give that clock a continuum unit; t_i remains a sweep count scaled by d_s/n_m .

Numerical (T-canonical, $\varepsilon_{\text{univ}}$): $T_{\text{avg}} = 35.345$ sweeps; $t_i^{\text{spatial}} = 0.6 \times 35.345 = 21.21$ sweeps; $t_i^{\text{matter}} = (5/3) \times 35.345 = 58.91$ sweeps; ratio 25/9 (the same spatial/matter ratio as $T(R_1)$, ZC81).

Classification: Exact Finding (recovery of an existing object; no new content, re-purposed as the bridge's upper half).

0.49 THM-ZC87-01 The Temporal Continuum Bridge

THM-ZC87-01: dt Is the Tick, Made Dense ($\varepsilon_0 \rightarrow 0$, Near-Formal)

Statement.

$$dt = \tau_{\text{sweep}}, \quad \text{continuum} \iff T_{\text{avg}}(\varepsilon_0) = \frac{2nK^*}{(n-1)\varepsilon_0} \xrightarrow{\varepsilon_0 \rightarrow 0} \infty. \quad (34)$$

The continuum time increment *is* the physical sweep tick τ_{sweep} . What makes time *smooth* (continuum) is that the number of ticks per renewal window, T_{avg} , diverges as $\varepsilon_0 \rightarrow 0$: the discrete tick-sequence $\{k \tau_{\text{sweep}}\}_{k \in \mathbb{N}}$ becomes dense relative to any fixed observable window.

Proof (near-formal).

Step 1 (One tick advances time by τ_{sweep}). A renewal window of physical duration D_R contains T_{avg} sweeps. The time advanced per sweep is D_R/T_{avg} , which is by definition τ_{sweep} (the duration of one sweep). Hence the elementary increment is $dt = \tau_{\text{sweep}}$ — nothing is resolved below one sweep, so no smaller increment is physical.

Step 2 (Tick-density diverges; transfer of THM-ZC51-01). $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0] \rightarrow \infty$ as $\varepsilon_0 \rightarrow 0$ (numerically $T_{\text{avg}}(10^{-3}) = 268.6$, $T_{\text{avg}}(10^{-4}) = 2686$, $T_{\text{avg}}(10^{-6}) = 2.69 \times 10^5$). THM-ZC51-01 established that CRF discrete observ-

ables converge to continuum values as $\varepsilon_0 \rightarrow 0$, with finite offset LEM-ZC51-02. Here the refining observable is the tick *density* per renewal: as it diverges, the step function built from $\{k \tau_{\text{sweep}}\}$ converges to a smooth time line.

Step 3 (Fixed-observable structure matches ZC51 exactly). ZC51 holds $\lambda_2 T_{\text{avg}} = (n - 1) \cdot 29/30$ fixed while $\lambda_2 \rightarrow 0$ and $T_{\text{avg}} \rightarrow \infty$. The time-axis analogue holds τ_{sweep} fixed (it is the single calibration constant) while the tick rate per window $\rightarrow \infty$. The fixed object is τ_{sweep} ; the diverging object is T_{avg} . No quantity is sent pathologically to a degenerate value.

Step 4 (One unknown in, one out). T_{avg} is derived and dimensionless; τ_{sweep} is the single uncalibrated duration, and $dt = \tau_{\text{sweep}}$ carries it directly. No second unknown is introduced (LEM-ZC87-01). $\square$

Why “Near-Formal”. Step 2 invokes the THM-ZC51-01 continuum-convergence result by transfer; a fully rigorous time-axis proof would carry a LEM-ZC51-02-type finite- ε_0 density correction, which is expected to be small and is not computed here. The *relation* $dt = \tau_{\text{sweep}}$ and the divergence of T_{avg} are exact.

What this closes. The *structural* half of GAP-ZC76-01: the sweep tick and the continuum dt are identified, and the continuum character is derived from tick density. What remains is a single calibration (τ_{sweep} in seconds), gated on external experiment — the gap is refined from “derive the bridge” to “measure one constant.”

Why $dt = \tau_{\text{sweep}}$ and not $\tau_{\text{sweep}}/T_{\text{avg}}$ (the v1 correction)

Version 1 of this paper wrote $dt = \lim \tau_{\text{sweep}}/T_{\text{avg}}$. That quantity is *sub-tick*: smaller than one sweep, hence below the resolution floor where no time is defined. The error was conflating the *duration of a renewal window* ($\tau_{\text{sweep}} T_{\text{avg}}$) with the *per-tick increment* (τ_{sweep}). The continuum does not come from shrinking the tick; it comes from packing more fixed-size ticks into each window ($T_{\text{avg}} \rightarrow \infty$). This is recorded as PM-ZC87-02 and is what closed GAP-ZC87-01.

0.50 LEM-ZC87-01 The Bridge Is Not Circular

LEM-ZC87-01: dt_{int} (absorbed) and dt_{em} (built dt -free) Are Distinct (R_0 , Exact)

Statement. The continuum dt inside DEF-RN02 and the emergent dt of THM-ZC87-01 are different objects; no feedback loop exists.

Proof.

Step 1 (dt_{int} is absorbed). In DEF-RN02, $T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$. By ZC30 (equating with $T(R_1) = (d_s/n_m)\varepsilon_0^2$ for spatial-type),

$$\Sigma_{\text{inv}} dt_{\text{int}} = \frac{d_s}{n_m} = 0.6.$$

The symbol dt_{int} never appears alone — only in the product $\Sigma_{\text{inv}} dt_{\text{int}}$, which is a pure CRF number. It is an internal bookkeeping infinitesimal, already eliminated.

Step 2 (dt_{em} is built dt -free). $dt_{\text{em}} = \tau_{\text{sweep}}$, the physical sweep tick, which is the single calibration constant of the cascade. It is not assembled from dt_{int} at all; it is an independent physical duration whose value is set (eventually) by external

calibration, while its *role* as the time increment follows from THM-ZC87-01. In particular it does not contain dt_{int} .

Step 3 (No loop). Since dt_{em} is assembled entirely from dt -free quantities and dt_{int} is absorbed before any rate law uses it, neither feeds the other. The bridge does not consume its own output. $\square$

Consequence. The bridge converts exactly *one* unknown (τ_{sweep}) into dt ; it creates no second unknown. This is why GAP-ZC76-01 reduces to a single calibration constant rather than expanding.

Classification: Exact Lemma.

0.51 FIND-ZC87-02 Gravitational Dilation as Sweep Slow-down

FIND-ZC87-02: Time Dilation Is a Slowdown of the δ -Resolution Rate (R_1 , Near-Formal — Gift, bonus)

Statement. Placing dt_{em} beside the Part A proper-time relation $d\tau/dt = 1/\sqrt{\mathcal{C}/\alpha}$: in the flat region $\mathcal{C} = \alpha$, $d\tau = dt_{\text{em}}$ (one sweep-window per proper-time unit). Near a mass, $\mathcal{C} \gg \alpha$, so $d\tau < dt_{\text{em}}$: fewer sweeps are resolved per unit proper time. Gravitational time dilation *is* the slowdown of the δ -resolution (sweep) rate in regions of high constraint density.

Reading. A clock deep in a gravitational well ticks slowly, in CRF, because the δ -cascade resolves fewer sweeps there per unit of the emergent dt . The metric factor $\sqrt{\mathcal{C}/\alpha}$ is the local sweep-rate suppression. This recasts a kinematic GR effect as a resolution-rate statement — and it is falsifiable: any setting where CRF predicts a sweep-rate change must match the measured time-dilation factor.

Status note. This is a *bonus* reading, registered separately from the core bridge. It rests on the Part A metric derivation (near-formal); if it were wrong, THM-ZC87-01 and LEM-ZC87-01 would stand unchanged. It is *not* load-bearing.

Classification: Near-Formal Finding (Gift, bonus).

0.52 Phase Misalignment and Open Gap

PM-ZC87-01: The Dirac-Number Probe That Fed Its Own Answer (pattern: tautology test)

The probe. While attacking GAP-ZC76-01, a probe computed

$$N_{\text{sweep}} = \frac{t_{\text{Hubble}}}{t_{\text{Planck}}} \approx 8.4 \times 10^{60},$$

and noted it matched the Dirac large number $\sim 10^{61}$. For a moment this looked like CRF *predicting* the Dirac number.

Why it failed. t_{Hubble} is the observed cosmic age and t_{Planck} an observed constant; their ratio is the ratio of two *measured* inputs. No CRF derivation occurred — the number was fed in and read back out. A search for any path to $10^{60.9}$ from CRF constants alone ($n, K^*, \mathcal{G}, \varphi, e, |\mathbb{Z}_{15}|$) returned only non-integer, non-structural

exponents: forcing a fit would be numerology, the pattern retracted in COR-ZC81-02/82-02.

What the failure revealed (the gift inside the scar). The audit forced a partition of cosmic quantities by predictability:

- **Constant** (τ_{sweep} , the tick): derivable in *structure* (Axiom 10), gated only on calibration.
- **Rate** ($\dot{G}/G = \mathcal{G}H_0$): derivable (FIND-ZC76-01).
- **Epoch** ($N_{\text{sweep}} \approx 10^{61}$, the accumulated sweep count): *not* derivable — it is a clock reading that grows every tick, not a structural constant. Its non-derivability is a feature: a theory that derives constants and rates should *not* output the present count.

This partition is precisely why THM-ZC87-01 closes the *structural* bridge but not the epoch: dt emerges, the rate is fixed, but the accumulated N_{sweep} remains an observer's clock reading.

Pattern. Tautology test — fed an observed quantity, read it back as derived. Caught by provenance audit (PWA-5).

Status: Scar. Preserved: it is what produced the constant/rate/epoch partition.

COR-ZC87-01 (was GAP-ZC87-01): Joint Limit Is Well-Defined — CLOSED

Resolution. The v1 worry — that the limit needed $\tau_{\text{sweep}} \rightarrow 0$ with $T_{\text{avg}} \rightarrow \infty$ on a constrained surface — dissolves under the corrected relation $dt = \tau_{\text{sweep}}$ (THM-ZC87-01). The limit holds τ_{sweep} *fixed* (it is the single calibration constant) and sends only the tick-density $T_{\text{avg}} \rightarrow \infty$. This is exactly the ZC51 fixed-observable structure: ZC51 fixes $\lambda_2 T_{\text{avg}}$ while $\lambda_2 \rightarrow 0$; ZC87 fixes τ_{sweep} while the per-window tick count $\rightarrow \infty$. No quantity is driven to a degenerate value, so the limit is trivially well-defined and yields the finite, nonzero increment $dt = \tau_{\text{sweep}}$.

Verification (four consistency tests, all pass).

1. Part A $d\tau/dt = 1/\sqrt{C/\alpha}$: flat region gives $d\tau = \tau_{\text{sweep}}$; near mass $d\tau < \tau_{\text{sweep}}$ (dilation, FIND-ZC87-02). Consistent.
2. Tick-density: $T_{\text{avg}}(\varepsilon_0) \rightarrow \infty$ packs fixed-size ticks densely; the step sequence $\{k\tau_{\text{sweep}}\}$ becomes smooth. Consistent.
3. Dimensions: $dt = \tau_{\text{sweep}}$ is time = time; $\tau_{\text{sweep}}/T_{\text{avg}}$ would be sub-tick (rejected).
4. ZC33: $t_i^{\text{phys}} = (d_s/n_m)T_{\text{avg}}\tau_{\text{sweep}}$ has per-tick increment τ_{sweep} . Consistent.

Classification: Corollary (gap closure, Near-Formal). A fully rigorous time-axis density-convergence proof (LEM-ZC51-02-type correction) would upgrade this to Exact; the structure is settled.

PM-ZC87-02: The Sub-Tick Formula (pattern: phase misalignment, wrong scaling layer)

The probe. Version 1 of this paper stated $dt = \lim_{\varepsilon_0 \rightarrow 0} \tau_{\text{sweep}}/T_{\text{avg}}$, reasoning that a renewal window of duration $\tau_{\text{sweep}}T_{\text{avg}}$ should be divided by its sweep count to give the increment.

Why it failed. $\tau_{\text{sweep}}/T_{\text{avg}}$ is *sub-tick*: smaller than one sweep, hence below the resolution floor where no time is defined. The error conflated the *renewal-window du-*

ration ($\tau_{\text{sweep}} T_{\text{avg}}$) with the *per-tick increment* (τ_{sweep}). Three trial fixed-observable choices all sent $dt \rightarrow 0$, exposing the mistake.

What the failure revealed. The continuum does not arise from shrinking the tick but from packing more fixed-size ticks per window ($T_{\text{avg}} \rightarrow \infty$). Correcting to $dt = \tau_{\text{sweep}}$ simultaneously closed GAP-ZC87-01 (now COR-ZC87-01): the confusing joint limit was an artifact of the wrong formula.

Pattern. Phase misalignment — wrong scaling layer (window-duration vs per-tick increment). The arithmetic was fine; the physical identification of dt was off by a factor T_{avg} .

Status: Scar. Preserved: it is what closed the gap.

GAP-ZC76-01: Refined to a Single Calibration

Refinement. With THM-ZC87-01 ($dt = \tau_{\text{sweep}}$) and COR-ZC87-01 (limit well-defined), GAP-ZC76-01 (GtRate-1, $\tau_{\text{sweep}} = t_{\text{Planck}}$) splits:

- **Structural bridge sweep** $\leftrightarrow dt$: CLOSED (THM-ZC87-01 + COR-ZC87-01, near-formal).
- **Calibration τ_{sweep} in seconds:** still GATED on external SNSPD timing. The gap is now exactly *one* constant.

Status: Refined (not closed). The external block is now a single-number calibration (τ_{sweep}), not a missing derivation.

0.53 Atomic Registry

Atomic Registry — ZC87

ID	Description	Status
FIND-ZC87-01	ZC33 sector physical time $t_i = (\Sigma_{\text{inv}} dt) T_{\text{avg}} = (d_s/n_m) T_{\text{avg}}$ is the bridge's upper half (sweep→clock); dt absorbed	Exact (recovery)
THM-ZC87-01	$dt = \tau_{\text{sweep}}$; continuum from tick-density $T_{\text{avg}} \rightarrow \infty$; transfers THM-ZC51-01 fixed-observable limit to the time axis; closes structural GAP-ZC76-01	Near-Formal
LEM-ZC87-01	dt_{int} (absorbed in d_s/n_m) $\neq dt_{\text{em}}$ (built dt -free); bridge not circular; one unknown in, one out	Exact
FIND-ZC87-02	gravitational dilation = sweep-rate slow-down ($d\tau/dt = 1/\sqrt{C/\alpha}$); testable; bonus, not load-bearing	Near-Formal (Gift)
PM-ZC87-01	$t_{\text{Hubble}}/t_{\text{Planck}} \sim 10^{61}$ mistaken for derivation; fed observed inputs; produced constant/rate/epoch partition	Scar
PM-ZC87-02	v1 wrote $dt = \tau_{\text{sweep}}/T_{\text{avg}}$ (sub-tick); corrected to $dt = \tau_{\text{sweep}}$; pattern: wrong scaling layer	Scar
COR-ZC87-01	(was GAP-ZC87-01) joint limit well-defined: τ_{sweep} fixed, tick-density $T_{\text{avg}} \rightarrow \infty$; four tests pass	Near-Formal
GAP-ZC76-01	refined: structural bridge closed; τ_{sweep} calibration gated on external	Refined

0.54 R-Layer Research Note

R-Layer Assignments for ZC87

Atomic unit	R-layer	Cost T	Source
FIND-ZC87-01	$R_0 \rightarrow R_1$	$(\Sigma_{\text{inv}} dt) T_{\text{avg}}$	ZC33 + ZC30
THM-ZC87-01	$R_0 \rightarrow R_1$	tick-density limit	ZC51 + ZC62
LEM-ZC87-01	R_0	algebraic	DEF-RN02 + ZC30
FIND-ZC87-02	R_1	metric	Part A ($d\tau/dt$)
COR-ZC87-01	$R_0 \rightarrow R_1$	limit closure	ZC51 transfer

Two-layer discipline (PWA-8). The entire paper lives on the $R_0 \rightarrow R_1$ crossing: R_0 supplies the discrete sweep (lower time), R_1 supplies the continuum dt (upper time). The bridge is the crossing itself. The load-bearing caution (LEM-ZC87-01) is keeping dt_{int} (an R_1 bookkeeping infinitesimal, absorbed) distinct from dt_{em} (the $R_0 \rightarrow R_1$ coarse-graining output). Conflating them would make the bridge circular.

0.55 Genealogy Map

How ZC87's Key Objects Trace Back

Branch A — The upper half (sweep $\rightarrow$ clock):

- **ZC30/FIND-ZC30-02:** $\Sigma_{\text{inv}} dt = d_s/n_m$ (dimensionless; spatial/matter 25/9).
- **ZC62/COR-ZC62-02:** $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$.
- **ZC33/DEF-ZC33-01:** $t_i = (\Sigma_{\text{inv}} dt)T_{\text{avg}}$; semigroup clock (THM-ZC33-02).
- $\rightarrow$ **ZC87/FIND-ZC87-01:** recognised as the bridge's upper half.

Branch B — The lower half (sweep $\rightarrow$ dt):

- **ZC51/THM-ZC51-01:** continuum limit $\varepsilon_0 \rightarrow 0$ + LEM-ZC51-02 correction.
- $\rightarrow$ **ZC87/THM-ZC87-01:** same machinery on the time axis; $dt = \tau_{\text{sweep}}$, continuum from tick-density $T_{\text{avg}} \rightarrow \infty$.

Branch C — Non-circularity:

- **DEF-RN02:** $T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$ (dt_{int} enters).
- **ZC30:** absorbs it into d_s/n_m .
- $\rightarrow$ **ZC87/LEM-ZC87-01:** $dt_{\text{int}} \neq dt_{\text{em}}$; no loop.

Branch D — The bonus, the gate, and the correction:

- **Part A:** $d\tau/dt = 1/\sqrt{\mathcal{C}/\alpha}$; $\mathcal{C}(r)$ continuum (#17).
- $\rightarrow$ **ZC87/FIND-ZC87-02:** dilation = sweep slowdown.
- **ZC76/GAP-ZC76-01** $\rightarrow$ refined to one calibration.
- **COR-ZC87-01** (was GAP-ZC87-01): joint limit closed once the v1 sub-tick formula was corrected (PM-ZC87-02) to $dt = \tau_{\text{sweep}}$.

Convergence at ZC87: Branch A (upper half, forgotten in ZC33) and Branch B (lower half, the ZC51 transfer) meet in the middle — the missing bridge between CRF's two times. Branch C guards the join against circularity; Branch D adds a testable dilation reading, refines the external gap to a single constant, and — through the corrected formula — closes the joint-limit question. ZC87's contribution: continuum time is no longer a primitive in CRF — it is the δ -sweep tick made dense, and the only thing still owed to experiment is the duration of one tick.

0.56 Closing Sentence

CRF kept two clocks: one that counted, one that flowed.

*The counting clock was built in ZC33 and set aside;
the flowing clock was borrowed, never grounded.*

*We first thought the flow was the tick divided by the count —
but that is smaller than a tick, and nothing lives below a tick.*

*The flow is the tick. What makes it smooth
is letting each window hold uncountably many of them.*

$$dt = \tau_{\text{sweep}}, \quad T_{\text{avg}} \xrightarrow{\varepsilon_0 \rightarrow 0} \infty.$$

What time is, CRF now says.

How long one tick lasts — that, still, the world must tell us.

Part LVI

Cross-Part Connection Map v17.15 (Part K)

Cross-Part Connection Map v17.15 — Reading Conventions

This map records, for each paper in Part K (ZC83–87): (1) **Backward Inheritance** — which objects from Parts A–J are used as input; (2) **Forward Corrections** — which prior-Part gaps are closed or refined by Part K; (3) **Cross-Part Phase Misalignments** — PMs that span more than one Part. Source files are never modified (No-Loss).

Notation: ✓ **CLOSED** = gap resolved. ★ **PROMOTED** = conjecture promoted to theorem. **PM-V17_15-EPSFAMILY:** local overrides $\backslash\text{epsEM} \rightarrow \varepsilon_{\text{EM}}$, $\backslash\text{epscan} \rightarrow \varepsilon_{\text{can}}$, $\backslash\text{epsSC} \rightarrow \varepsilon_{\text{SC}}$, $\backslash\text{epsS} \rightarrow \varepsilon_{\text{strong}}$ (bare sector forms, Style Guide v3.5); extends PM-V17_13-EPSEM of Part J. Chain and sources unmodified.

ε FLOOR VALUES — VERIFIED CONSISTENT — DO NOT “RECONCILE” (read before editing any ε line)

If you (human or AI) see the near-equal floor values 0.007550, 0.007599, 0.007605 appearing alongside **COR-ZC67-01** (and now **FIND-ZC83-01**) and conclude they contradict — **they do not**. This was audited line-by-line across ZC66–87 and is the single most common trap in this corpus.

COR-ZC67-01 / FIND-ZC83-01 are ONE exact identity, $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_{\text{W}}} = \varepsilon_{\text{base}}$. It evaluates to a *different number on each layer’s input pair*:

- ZC66/67 defined pair (ε_{can} base) $\rightarrow 0.007550 = \varepsilon_{\text{can}}$
- ZC70/81/82/83 defined pair ($\varepsilon_{\text{univ}}$ base) $\rightarrow 0.007599 = \varepsilon_{\text{univ}}$ (the R_0 -layer GM; FIND-ZC83-01)
- observed/empirical pair (the two FIND-ZC65-02 bridge floors) $\rightarrow 0.007605 = \varepsilon_{\text{SC}}$ — the R_{max} -layer GM (FIND-ZC85-01); its +0.08% above $\varepsilon_{\text{univ}}$ is the geometric mean of the two running shifts, gated on and then released by GAP-ZC81-01 (ZC86/FIND-ZC86-01)

These three are *intentionally distinct* (different input pairs at different R -layers), not rounding drift and not a contradiction. ZC83–87 add the layer interpretation but change *no* value.

DIRECTIVE. Do not change any ε value, do not “reconcile” them, do not open an audit, and do not edit the COR-ZC67-01, FIND-ZC83-01, or FIND-ZC85-01 lines to fix a perceived conflict. If a conflict still seems present, you are reading one identity as if it were three values — re-read this box, not the papers. (ZC66–69 audited and correct; ZC81–87 correct.)

0.57 Connection Records for ZC83

ZC83 — Backward Inheritance from Parts A–J

- ZC67/LEM-ZC67-01: sector-floor formula (source of the ordering)
- ZC67/COR-ZC67-01: $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_W}$ geometric-mean structure
- ZC69/COR-ZC69-02: $\varepsilon_{\text{univ}}$ exact closed form
- ZC70/THM-ZC70-01: $\varepsilon_{\text{EM}}, \varepsilon_W$ exact values
- ZC71/THM-ZC71-01: two-layer separation (gap irreducibility)
- ZC79/THM-ZC79-01: ε_{can} exact
- ZC57/THM-ZC57-01: sector-weight ratio $r = C_W/C_{\text{EM}}$

ZC83 — Forward Corrections / Gaps

- Floor lattice ordering $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_W$: promoted to forced theorem (THM-ZC83-01) from prior numerical observation.
- FIND-ZC83-01: $\sqrt{\varepsilon_{\text{EM}}\varepsilon_W} = \varepsilon_{\text{univ}}$ exact at the R_0 layer (gift).
- GAP-ZC83-01 (ε_{SC} ontology): **opened** here; closed in Part LIV.

0.58 Connection Records for ZC84

ZC84 — Backward Inheritance from Parts A–J

- ZC20/THM-ZC20-01: $\sin^2 \theta_W = 1 - \ln 8 / \ln 15$
- ZC29/THM-ZC29-01: $\alpha_i = \varphi(d_i)/15 \cdot \Gamma_{\text{CRF}}$
- ZC72/THM-ZC72-01: $\alpha_{\text{grav}}(R_0) = \Gamma_{\text{CRF}}/15$
- ZC74/THM-ZC74-01: $\mathbb{Z}_{15}$ unique squarefree semiprime
- ZC75/THM-ZC75-01: $\alpha_{\text{grav}} \cdot |Z_{15}| = \Gamma_{\text{CRF}}$
- ZC80/THM-ZC80-01: 7 EW observables at R_0 , zero free params
- ZC81/THM-ZC81-01, FIND-ZC81-01: $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$; mirror symmetry
- ZC82/THM-ZC82-01: $w_{\text{DE}} + 3\Delta\Gamma = \Gamma_{\text{CRF}}$
- ZC83/THM-ZC83-01, FIND-ZC83-01: ordering and GM identity

ZC84 — Forward Corrections / Gaps

- THM-ZC84-01: eight prior results assembled into one synthesis (no new derivation — see PM-ZC84-SCOPE-01).
- COR-ZC84-01: four-sector falsification table, zero free params.
- Does not close GAP-ZC83-01, GAP-ZC81-01, or GAP-ZC76-01.

0.59 Connection Records for ZC85

ZC85 — Backward Inheritance from Parts A–J

- ZC65/FIND-ZC65-02: two per-observable bridge floors ($\varepsilon_0^\alpha, \varepsilon_0^{\text{sw}}$)
- ZC67/COR-ZC67-01, LEM-ZC67-01: GM structure
- ZC68/THM-ZC68-01b: ε_{SC} observed bracket
- ZC69/COR-ZC69-02: $\varepsilon_{\text{univ}}$ exact

- ZC70/THM-ZC70-01: $\varepsilon_{\text{EM}}, \varepsilon_W$ exact
- ZC81/GAP-ZC81-01: Strong/EW running bridge (the gate)
- ZC83/GAP-ZC83-01, FIND-ZC83-01: the gap refined; R_0 -layer GM

ZC85 — Forward Corrections / Gaps

- GAP-ZC83-01 (ε_{SC} ontology): **structurally** ✓ **CLOSED** (FIND-ZC85-01: ε_{SC} = GM of the two bridge floors).
- GAP-ZC83-01 magnitude: refined to **gated** on GAP-ZC81-01 (COR-ZC85-01), no longer an independent open problem.

0.60 Connection Records for ZC86

ZC86 — Backward Inheritance from Parts A–J

- ZC65/THM-ZC65-01, FIND-ZC65-01/02, LEM-ZC65-01: EM bridge
- ZC30/FIND-ZC30-02: spatial vs matter-type sector class
- ZC73/THM-ZC73-01: Γ ladder; Γ_{strong} at $k = 1$
- ZC81/THM-ZC81-01: $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$; LEM-ZC81-01: T_{R_1} crossing cost; GAP-ZC81-01 (the gate it closes)
- ZC85/COR-ZC85-01: ε_{SC} magnitude gated on GAP-ZC81-01

ZC86 — Forward Corrections / Gap Closures

- GAP-ZC81-01 (Strong running bridge): ✓ **CLOSED** (THM-ZC86-01, near-formal; bridge form and magnitude = EM).
- GAP-ZC83-01 magnitude (via COR-ZC85-01 gate): ✓ **CLOSED** (FIND-ZC86-01: the running floor layer has exactly two floors; ε_{SC} = their GM).
- LEM-ZC86-01: confinement decoupling — confinement lives in Γ , not in α (removes the only obstruction).

0.61 Connection Records for ZC87

ZC87 — Backward Inheritance from Parts A–J

- ZC33/DEF-ZC33-01: sector physical time (the recovered upper half); THM-ZC33-02: semigroup clock
- ZC51/THM-ZC51-01: continuum-limit machinery ($\varepsilon_0 \rightarrow 0$)
- ZC30/FIND-ZC30-02: $\Sigma_{\text{inv}} dt = d_s/n_m$; sector class
- ZC62/COR-ZC62-02: $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon] = eF/K^*$
- ZC76/FIND-ZC76-01: $\mathcal{G}H_0$ rate; GAP-ZC76-01 (GtRate-1)
- Part A: the $\Sigma_{\text{inv}} dt$ relation (FIND-ZC87-02 bonus)

ZC87 — Forward Corrections / Gap Closures

- GAP-ZC87-01 (joint-limit well-definedness): ✓ **CLOSED** in v2 (τ_{sweep} fixed, $T_{\text{avg}} \rightarrow \infty$).

- GAP-ZC76-01 (τ_{sweep}): structural sweep $\leftrightarrow$ dt bridge ✓ **CLOSED** (THM-ZC87-01, near-formal); seconds-calibration remains externally blocked.

0.62 Cross-Part Phase Misalignment Registry (Part K)

Part K — Phase Misalignments Preserved as Scar Tissue

- **PM-ZC83-01** — the “is ε_{SC} the arithmetic mean of the two floors?” probe. Pattern: false-pattern test; ruled out exactly (AM $> \varepsilon_{\text{univ}}$, FIND-ZC83-02).
- **PM-ZC84-SCOPE-01** — synthesis paper: every identity in THM-ZC84-01 is inherited from a prior theorem; none is newly derived. Pattern: scope honesty.
- **PM-ZC85-01** — “ $\varepsilon_{\text{SC}} = \text{GM of the bridge pair}$ ” is a restatement of FIND-ZC65-02 + COR-ZC67-01, not a new derivation. Pattern: tautology test (PWA-5).
- **PM-ZC86-01** — universal-coupling propagation path: propagating the bridge through an effective universal coupling G^{eff} gave 0.15%/0.41% errors. Failed; preserved as the path that lit the correct route. Pattern: probe malformed.
- **PM-ZC87-01** — the Dirac-large-number probe: $N_{\text{sweep}} = t_{\text{Hubble}}/t_{\text{Planck}} \sim 10^{61}$ read as a CRF “prediction” of the Dirac number, when it was the ratio of two observed inputs feeding itself. Pattern: circular anchor.
- **PM-ZC87-02** — v1’s $dt = \tau_{\text{sweep}}/T_{\text{avg}}$: a sub-tick quantity below one sweep; corrected in v2 by the fixed-observable probe ($dt = \tau_{\text{sweep}}$, density $T_{\text{avg}} \rightarrow \infty$). Pattern: definition drift.